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CUSTOMER EDUCATION SERVICES

Design Compiler Workshop

Lab Guide

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1 Setup and Synthesis Flow

Learning Objectives

After completing this lab, you should be able to:

- Describe the contents of each of the three DC setup files used in this lab.
- Update the `common_setup.tcl` file to fully specify the logic and physical library and technology files
- Explore the schematic view in *Design Vision*
- Take a design through the basic synthesis steps in Topographical mode and generate reports
- Visit *SolvNet* to browse the user manual for Design Vision

Lab Duration: 60 minutes

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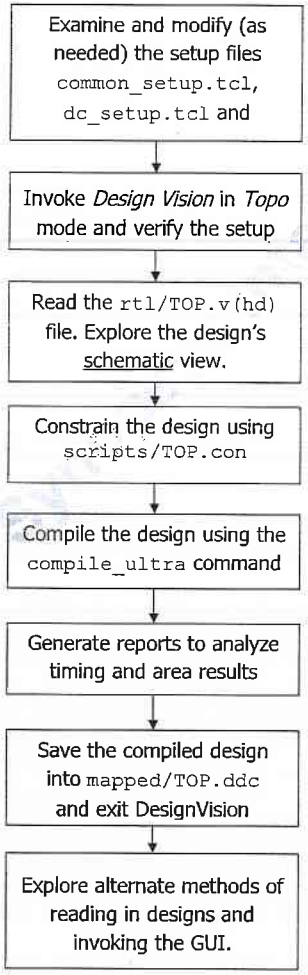
Lab Flow

Follow the detailed step-by-step **Lab Instructions** on the following pages to perform the steps highlighted in this flow:



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Lab Instructions

Task 1. Examine and Modify the Setup Files

1. Make the `lab1` directory as your working directory and list the following files:

```
UNIX% cd lab1
UNIX% ls -al .synopsys*
UNIX% ls -al *setup*
```

   You are provided with a `.synopsys_dc.setup` file. It defines aliases and sources two other setup files: `common_setup.tcl` and `dc_setup.tcl`.

   The `common_setup.tcl` file contains user defined variables (in UPPER CASE letters) which specify technology file and directory names. These variables are used in `dc_setup.tcl`.

   The `dc_setup.tcl` file executes commands to load the necessary logical and physical technology data, using variables from `common_setup.tcl`.

2. You will NOT have to modify the `.synopsys_dc.setup` or `dc_setup.tcl` files for this lab. However, verify that the `dc_setup.tcl` file contains the "*" in the `link_library`.

3. Using a text editor of your choice, edit `common_setup.tcl` and incorporate any missing information from the following table:

User-defined variable	Directory or File Names

`ADDITIONAL_SEARCH_PATH`
Additional `search_path` directories for logic (db) libraries, design files, and scripts
	`../ref/libs/mw_lib/sc/LM`
	`./rtl`
	`./scripts`

`TARGET_LIBRARY_FILES` Logical Technology Library file	`sc_max.db` (located in the LM view of the Milkyway reference library)
`SYMBOL_LIBRARY_FILES` Symbol library file	`sc.sdb` (located in the LM view of the Milkyway reference library)

This table is continued on the next page.

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User-defined variable	Directory or File Names
`MW_DESIGN_LIB` Milkyway Design Library Name	`TOP_LIB` (user-defined name)
`MW_REFERENCE_LIB_DIRS` Milkyway reference libraries (standard/macro/pad cells)	`../ref/libs/mw_lib/sc`
`TECH_FILE` Physical Technology file	`../ref/libs/tech/cb13_6m.tf`
`TLUPLUS_MAX_FILE` Max TLUPlus file	`../ref/libs/tlup/cb13_6m_max.tluplus`
`MAP_FILE` TLUPlus Layer Mapping file	`../ref/libs/tlup/cb13_6m.map`

Task 2. Invoke Design Vision

1. Invoke Design Vision in *Topographical mode* from the `lab1` directory. Recall that this step automatically sources the `.synopsys_dc.setup` file.

```
unix% pwd
unix% design_vision -topo
```

Note: If you get a `MWUI-004` error message, "`Library 'TOP_LIB' already exists`", this indicates that the `TOP_LIB` *Milkyway* Design Library directory, which will be created later in this lab, already exists. If you are starting over because of a problem with the lab, you should `exit` the tool, delete the `TOP_LIB` unix directory, and re-invoke *Design Vision*. Otherwise, you can ignore the message.

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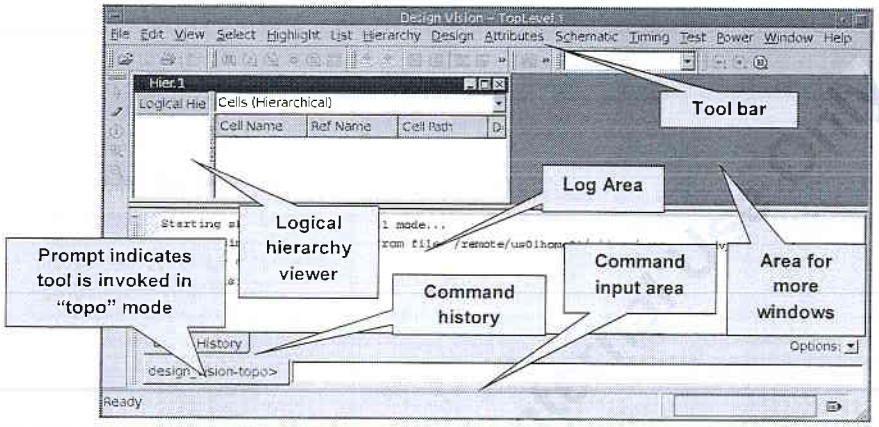

2. You can confirm that the tool is invoked in "Topographical mode" by noticing `-topo` in the command prompt, or by looking for the startup message that indicates "*Starting shell in Topographical mode...*".

3. View the ***Log Area*** at the bottom of the GUI.

   This area displays all the executed commands, their results, and shows any error messages.

   Scroll up to see the log messages.

   You should see the values of the key "`Library Setting`" variables echoed here, followed by the "`Initializing gui preferences`" message, and the "`gui_start`" command, which invokes the GUI.

4. Choose menu **File → Setup...** and verify that the libraries are set up correctly:

   Question 1. What is the *Link library*?

   .....

   Question 2. What is the *Target library*?

   .....

   Question 3. What is the *Symbol library*?

   .....

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Note: If the libraries are called `your_library.db (or "--")`, (instead of `sc_max.db`) this indicates you invoked DC from the wrong Unix directory. Exit the GUI (**File → Exit → OK**, or type **exit** at the command prompt, and choose **OK** when prompted). Make sure your current directory is `lab1` and re-invoke Design Vision.

Note: Check your answers against the **Answers/Solutions** section at the end of this lab, and fix your setup files accordingly. If you are stuck, compare your setup file with that provided in the ***.solutions*** directory.
If you edited any of the the "setup" files. (`i.e.`, `synopsys_dc.setup`, `common_setup.tcl` or `dc_setup.tcl` files) it is recommended that you exit Design Vision and re-invoke it.

5. From **File→Setup...** select the icon to the right of the ***Search path*** field. This opens up the *Set Search Path* dialog, which shows an expanded list of each search path directory. Verify that the first five entries at the top are as follows – these are the default `search_path` directories:

```
.
/<tool_installation_directory>/libraries/syn
/<tool_installation_directory>/minpower/syn
/<tool_installation_directory>/dw/syn_ver
/<tool_installation_directory>/dw/sim_ver
```

   Question 4. What user directories have been added to the *Search path*?

   .....

6. Click **Cancel** twice to close the *Set Search Path* and *Application Setup* dialogs.

7. In the original *Unix* window from which you invoked *Design Vision* (the shell interface), type the following commands at the DC prompt. This is another way to confirm variable settings, as well as user-defined and default aliases.

   Note: Command line editing allows for command, option, variable and file completion. Type a few letters and then hit the **[Tab]** key.

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Note: If your <Backspace> key deletes entire words, use Ctrl-H to back-space one character at a time.

```
printvar target_library
printvar link_library
printvar search_path
alias
```

8. Check the consistency between the logical and physical libraries:

```
check_library
```

Notice that the check reports that there are 4 cells missing in the logic library. There is a table listing the missing cells. They are "feedthrough" and "tap" cells, which are required in the physical layout, but not in the logic design (also known as "physical-only" cells). The warning can therefore be ignored.

9. Check the consistency between the *TLUPus* and the *Technology* files:

```
check_tlu_plus_files
```

You should see that all three checks "`Passed!`".

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Task 3. Read the Design into DC Memory

Design Compiler can read *VHDL*, *Verilog*, as well as *SystemVerilog RTL* files.

1. Click on the **Read...** button at the top left of the GUI (or **File → Read**).
2. In the dialog box that appears, double-click on the directory `rtl`, and then again on `TOP.vhd` or `TOP.v`.

   In the "Logical Hierarchy" window on the left side of the GUI (you may need to widen the window), there is now an icon for `TOP`, which is the top-level *design* name. There are also icons for the lower-level instances or *cells*: `I_FSM`, `I_DECODE`, and `I_COUNT`.
3. Select `TOP` (single click with left mouse button), and look at the lower right corner of the GUI window to verify the selection.
   You should see **Design: TOP**. This ensures that your *current design* is properly set to the top-level design.
4. Select **File → Link Design → OK** to link the design and resolve all references. You should not see any warning or error messages in the *Log Area*.
5. Save the unmapped design in *ddc* format. Type the following in the *Command Input Area* or in the Unix window in which you invoked Design Vision:

   Note: Command line editing allows for command, option, variable and file completion. Type a few letters and then hit the **[Tab]** key.

```
write_file -hier -f ddc -out unmapped/TOP.ddc
```

6. Type the following non-GUI *dc_shell* commands to see a list of designs and libraries in memory:

```
list_designs
list_libs
```


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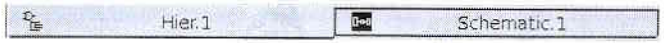
Task 4. Explore the Schematic View

1. Make sure that the lower right corner still shows that **Design: TOP** is selected. If not, select `TOP` with a single click of the left mouse button in the *Logical Hierarchy* window.

2. Select the *schematic view* by clicking the (**Create Schematic of Selected Objects**) icon in the tool bar. You will see the top level interface diagram (known as a "*Symbol View*" in earlier tool versions) of the block called *TOP*, with its input and output ports. The diagram is labelled `Schematic.1` in the upper left corner of the new window. Double click on the diagram. The schematic of `TOP` will now expand into a block diagram containing the instantiations of `FSM`, `DECODE` and `COUNT`.

   You now have two windows open in the GUI: The Hierarchical, and Schematic view windows. Maximize the schematic window. The Logical Hierarchy window is also maximized, but is "behind" the Schematic window that you maximized. You can bring diferent windows to the foreground by selecting the appropriate tab below the view window. Press the **[F]** key to "fit" the schematic view to the full window.

3. Minimize the view windows or select the left-most *Hierarchy* tab (or using the **Hierarchy → New Logical Hierarchy View**) to make the *Logical Hierarchy* window visible.

4. Explore `TOP` by visiting the **Schematic View** of the various subdesigns by selecting each design in the **Logical Hierarchy** window and clicking on the icon.

   Because you have not compiled these designs yet, you will <u>not</u> see gates from the target technology library. You will see *GTECH* components. *GTECH* components are generic Boolean gates and registers that represent the generic, non-technology specific functionality of a design.

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Task 5. Explore the Mouse Functions

1. **Click and hold** the **right mouse button** in a **schematic view** to see the available mouse functions.

2. Either select **Zoom Fit All** with the left mouse button, or press the **[F]** key, to maximize the view. Now either repeat **Step 1** and select **Zoom In Tool**, or press **[Z]**. With the left mouse button click and drag the rectangular area you want to zoom into. Press the **[ESC]** key to exit out of the ZOOM mode.

3. Repeat **Step 1** and select **Pan Tool**, or use the **Up/Down/Left/Right arrow** keys to pan around a zoomed-in view.

4. Return to **Zoom Fit All [F]** or **Zoom Out Tool [-]** by using the appropriate mouse function.

5. A quicker way to zoom in and out is using "strokes". Press and hold the middle mouse button on the lower left corner of the rectangle you want to zoom into, then move the mouse while still pressed to the upper right corner. Only then release the middle mouse button. You just performed a zoom in gesture or stroke. By pressing the middle mouse button in place, a menu will appear with the defined strokes. Experiment with some more strokes. If your mouse has a middle scroll button, you can also zoom in and out with it.

6. Select or open a *Schematic* view of `TOP`.

7. This time switch to the Schematic View of `DECODE` by double clicking on the green block labelled `I_DECODE` in the `TOP` schematic.

   Note the cell name `I_DECODE` in the lower right corner of Design Vision. This signifies that `I_DECODE` is the **Current Instance**.

   Go back to the `TOP` schematic view by clicking on the **up-arrow** button from the menu (top banner).

   Repeat this step for `I_FSM`, and `I_COUNT`.

8. To select multiple objects, experiment with using your **left mouse button** and the **CTRL key**. Use the left mouse button to select the first object, then **left mouse button and CTRL key** to select additional objects. *Selected objects are highlighted in white.* Click in any black area to un-select the selected objects.

Recall the Basic Steps in Synthesis Flow

The four steps after "read" will be performed in the upcoming tasks:

- Read and translate RTL code (`read_vhdl/read_verilog`)
- Constrain the design (`source` a constraints file)
- Synthesize the design (`compile_ultra`)

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- Generate reports (`report_*`)
- Save the resulting netlist (`write_file, write_icc2_files`)

Task 6. Constrain TOP with a Script file

1. Open the **Schematic** view for `TOP`.

   Note: If your *Logical Hierarchy* window is closed, re-open it by selecting **Hierarchy → New Logical Hierarchy View**

2. Type the following at the command prompt on the bottom of the Design Vision window. Remember to take advantage of command completion by hitting the tab key.

```
source TOP.con
```

   Note: If the `source` command gives an error message, make sure that the *./scripts* directory has been appended to the `search_path` variable in the 3 setup files, or, type `source scripts/TOP.con` if you do not want to re-invoke the Design Vision.

   This will execute a script file, which constrains the `TOP` design. You will not be able to "see" the constraints in the view window, but they are there. In upcoming labs you will learn how to generate reports to verify constraints that have been applied to a design.

Task 7. Compile or Map to Vendor-Specific Gates

1. To compile the design, type the following command at the command prompt on the bottom of the Design Vision window:

```
compile_ultra
```

   Monitor the log as *compile* progresses. You will see various tables for the different optimization phases of compile. The "AREA" column indicates the design size. The "WORST NEG SLACK" column indicates by how much the critical or worst path in the design is violating, relative to its constraint (Actual delay – Expected delay). The "TOTAL SETUP COST" is the sum of all the violating path slacks for setup timing. When the optimization reaches a point of "diminishing returns", or, the slack numbers reach zero, which means that there are no violating timing paths in the design, `compile_ultra` ends.

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2. Use the right mouse zoom functions, the [Z] and [F] keys, the middle mouse button "strokes", or the buttons on the command bar to explore the **Schematic View** of TOP.
   Note that `compile_ultra` has automatically flattened or "ungrouped" the sub-designs `DECODE`, `FSM` and `COUNT`, to generate optimal timing and area results (you will learn about "auto ungrouping" later in this course).
   You will now see gates from the target technology library – no longer GTECH gates.

Task 8. Generate Reports and Analyze Timing

1. Go to the **Schematic View** of `TOP`.
2. At the `design_vision-topo>` prompt type:

```
rc
```

   `rc` is an alias that was specified in the `.synopsys_dc.setup` file. It executes the following command: `report_constraint -all_violators`

   Note: Use the shell window behind the GUI, to view the report.

   This report lists a summary of all constraint violations. You should see several "`max_delay/setup`" violations.

   You can also get more detailed timing path information by generating a timing report:

```
rt
```

   By default, `report_timing` shows the timing of the critical path.

   Record the following *Worst Timing Violation*:

   `Slack (VIOLATED):` _____

   Note: You don't need to worry about any timing violation for this lab, you will learn to analyze timing violation and choose strategies to improve timing later in this course.

3. Generate an area report, **`ra`**, and record the following:

   `Total cell area:` _____

4. Locate the histogram buttons at the top of the window. Hover your mouse over the middle button – a "tool hint" should display "*Create endpoint slack histogram*".

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5. Click on the **Create endpoint slack histogram** button, and in the dialog box that appears, select **OK**.

   You will see a histogram window displaying several "bins". Each bin represents a number of timing paths to endpoints (output ports or register input pins). A "green" bin indicates that all timing paths within that bin meet timing. A "red" bin indicates violating paths.

   When a bin is selected, it turns "yellow", and some details of the paths are listed in the right section of the dialog box.

6. Select the "red" violating bin (left mouse button), and on the right side, select one of the endpoints (left mouse button).

   You can show the path to the selected endpoint in the schematic, as follows:

   First select the left-most "**Create Schematic of Selected Objects**" button – this shows the endpoint of the violating path – a register. Now select the middle "**Add Paths to Path Schematic**" button – this adds the path leading up to the endpoint. You can optionally select the right-most "**Add Fanin/Fanout to Path Schematic**" button to include fan-ins or fan-outs of specific pins (a "green arrow") along the path. Alternatively, for this optional step, you can click the right mouse button and select *Add Logic → Fanin/Fanout*.

Task 9. Save the Optimized Design

After compile, or after any major steps, it is advisable to save the design. The native Design Compiler format is *ddc*, both for unmapped or mapped (compiled) designs.

1. Go back to the Schematic View of `TOP`.
2. Choose menu **File → Save As.**
3. Double click on the `mapped` directory.
4. Enter `TOP.ddc` in the *File name* field.
5. Verify that the **Save all designs in hierarchy** button is selected. This will save the entire design hierarchy into a single (`.ddc`) file.
6. Click **Open.**

   You just saved the gate-level netlist (the entire hierarchy) in 'ddc' format under the `mapped` directory. You can verify that the file was created by entering '`ls mapped`'.

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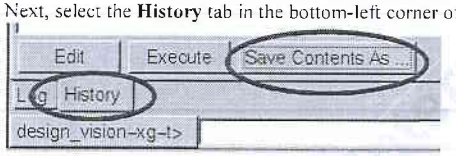
7. Save the gate-level design for IC Compiler II using the command:

```
write_icc2_files -output ./mapped/TOP_icc2
```

   Note: You can ignore the Error message that "`ScanDEF generation cannot proceed`": This is because we have not inserted scan chains yet (which uses a CTL model).

   Note: Use the "`-force`" option if the "`TOP_icc2`" directory already exists.

8. Next, select the **History** tab in the bottom-left corner of the GUI.



9. Save the command history by highlighting all the commands with the left mouse button, and then selecting the button "**Save Contents As**" and specifying the file name `dc.tcl` into the `scripts` directory.

Task 10. Remove Designs and Exit Design Vision

1. Remove all designs from DC memory:

```
fr
list_designs
```

   Notice that all the icons in Design Vision have been deleted. The "*fr*" alias executes the following command: `remove_design -designs`.

2. List a history of all commands executed since invoking Design Vision:

```
h ; # Alias for "history"
```

   These executed commands are automatically logged in the `command.log` file that has been created in your `lab1` project working directory. If you unintentionally exited out of Design Vision, you could recreate everything you did up to that point by doing the following:

   - Copy and rename the `command.log` file, and edit the copy to remove any "`exit`" or "`quit`" command at the end
   - Execute the log file:
     `UNIX% design_vision -topo -f command_copy.log`

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- Alternatively, you can use the `dc.tcl` file that you saved previously:
  `UNIX% design_vision -topo -f scripts/dc.tcl`

3. Exit from Design Vision. Use the menu sequence **File → Exit → OK**, or type `exit` at the command prompt, and choose **OK** when prompted.

4. Now try an alternate way to invoke the *DesignVision* GUI, by launching the interactive DC shell and then invoking the GUI:

```
UNIX% dc_shell -topo
dc_shell-topo> start_gui (or gui_start)
```

Note: For this step you can ignore the Error message:
`Library 'TOP_LIB' already exists.`

5. You can return to the shell mode either from the *Design Vision* GUI: **File → Close GUI**, or from the command line:

```
design_vision-topo> stop_gui (or gui_stop)
```

6. **Exit** the DC shell.

Task 11. (OPTIONAL) Browse Documentation on *SolvNet*

In this task we will pretend that we want to learn how to use Design Vision to print a design schematic. We will browse the on-line documentation to find the needed information. You will need your *SolvNet* ID and password for this task.

1. Open a web browser, enter the URL ***solvnet.synopsys.com***, and log on to *SolvNet* (the link to *SolvNet* is also available from the Synopsys home page):

```
unix% firefox &
```

2. You should see several "buttons" at the top of the main page, titled "Documentation", "Support", "Downloads", "Training" etc. Click on "**Documentation**".

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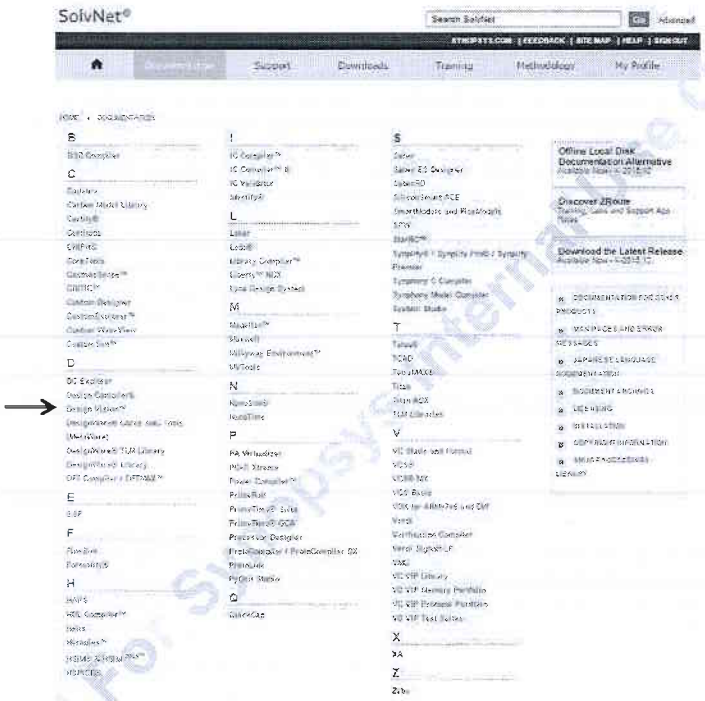
3. From the resulting "Documentation" product list, select **Design Vision**.



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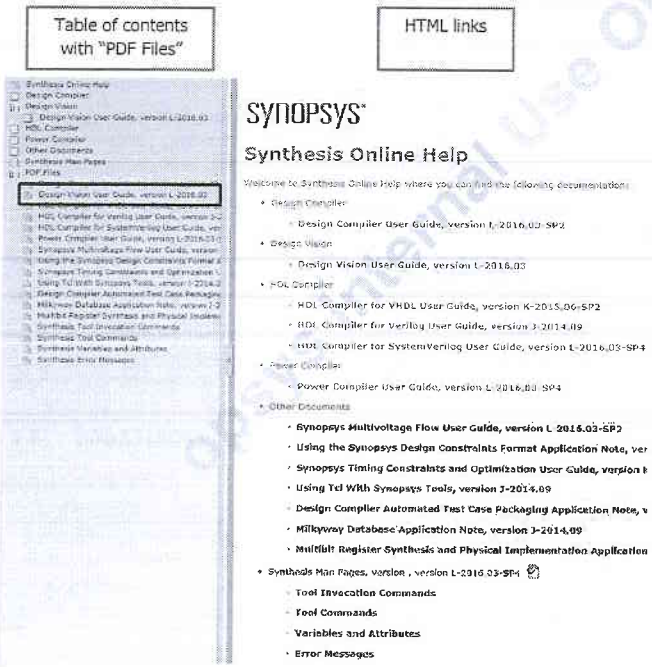
4. From the left *Contents* tab, expand "**PDF Files**" and open the *Design Vision User Guide*.



Note: Alternatively, open the *HTML* version by clicking the equivalent link on the right.

5. Once the user guide is open, go to the chapter on "**Performing Basic Tasks**", and then to "**Saving an Image of a Window or View**".

   You do not need to save anything - this is just one example of how to use the online documentation.

6. Exit the web browser.

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Task 12. (OPTIONAL) Using `analyze` and `elaborate` to read in an HDL design

For this task, you will invoke the tool in the *dc_shell* mode, and you will learn some additional ways to read in HDL files.

1. Invoke DC shell in Topo mode from the `lab1` directory:

```
unix% pwd
unix% dc_shell -topo
```

   There are several ways you can read a design into Design Compiler. In the beginning of this lab you used **File → Read** in *Design Vision*, which invokes the `read_verilog/vhdl` command.

2. Read a Verilog file in the *dc_shell* environment:

```
read_verilog ./rtl/TOP.v
current_design TOP
link
```

Note: You can read *VHDL* files using `read_vhdl` and *System Verilog* files using `read_sverilog`.

3. Remove all the designs from the DC memory:

```
fr
```

4. Another method to read a design is to use `analyze/elaborate`. The `analyze` command checks the syntax and synthesizability of the HDL. It also creates a directory called `analyzed`, by default, in which it stores an intermediate binary version of the design. The user can optionally change the directory in which these intermediate files are stored using the `define_design_lib` command (shown below). The `elaborate` command sets the "current design" to the specified design name (usually the top-level *module* or *entity*), links the design hierarchy, and translates the analyzed results into *ddc*. HDL designs containing parameters can be read in as templates, so that parameters may be redefined during the elaboration stage – an advantage over the `read` command which does not allow this:

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```
# Instead of the default "analyzed", write out the
# intermediate files in the directory called "work"
file mkdir ./work
define_design_lib WORK -path ./work
analyze -format verilog -library WORK ./rtl/TOP.v
elaborate TOP
```

Another method to read a design uses a *VCS* style option of the "analyze" command (*VCS* is Synopsys' functional verification tool). This is practical when you have *Verilog* files in different directories and do not know all the sub-design files required to link the design correctly. Here you need to provide only the top module *Verilog* file and `elaborate` will loads the other required *Verilog* files by searching specified directory paths.

5. Read the file using the "`-vcs`" option:

```
fr ;# Remove all the designs from DC memory
analyze -vcs " -verilog -y ./rtl +libext+.v " \
./rtl/TOP.v
elaborate TOP
```

Note: Currently `-vcs` options is supported only for *Verilog* and *System Verilog* files, and not for *VHDL*.

You have completed the "Setup and Synthesis Flow" lab of the Design Compiler Workshop.

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Answers / Solutions

Question 1. What is the *Link library*?

`* sc_max.db`

Question 2. What is the *Target library*?

`sc_max.db`

Question 3. What is the *Symbol library*?

`sc.sdb`

Question 4. What user directories have been added to the *Search path*?

```
../ref/libs/mw_lib/sc/LM
./rtl
./scripts
```


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2 Design and Library Objects

There is NO LAB for this unit

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3 Timing Constraints

Learning Objectives

After completing this lab you should be able to:

- Determine the unit of time used in the target library
- Create a Design Compiler timing constraints file based on a provided schematic and specification
- Verify the syntax of the constraints prior to applying them to a design
- Apply the constraints to a design
- Validate the completeness and correctness of the applied constraints

Lab Duration: 80 minutes

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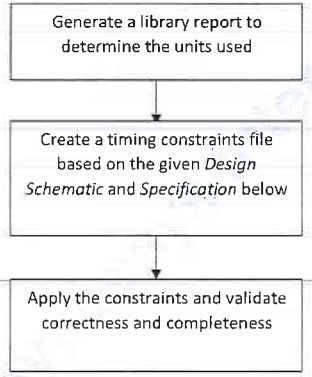
Lab Flow

Follow the step-by-step **Lab Instructions** on the following pages to perform the three tasks shown in the lab flow below. Refer to the **Design Schematic** and **Design Specification** sections are needed.



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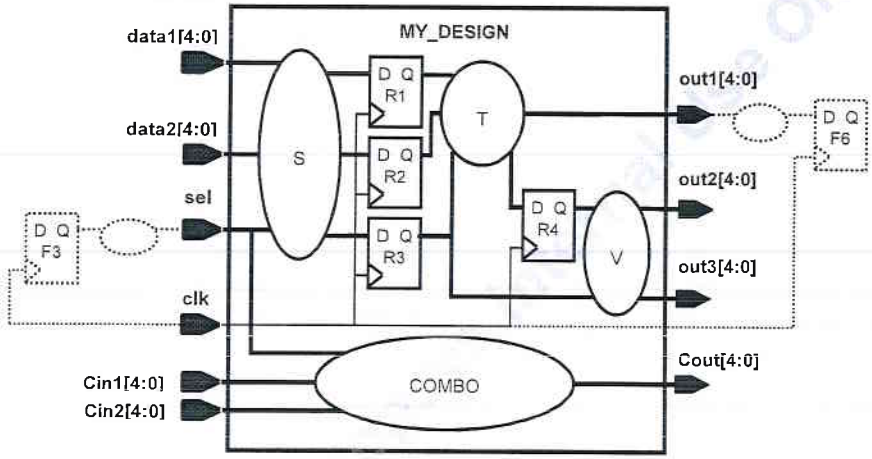
Design Schematic



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Design Specification

Hint: **Read carefully!** Some of the specifications are described in "non-DC" language or terms, requiring translation and calculation to derive the DC constraints.

Clock Definition	1. Clock `clk` has a frequency of **333.3⅓ Mhz.** 2. The maximum external clock generator delay to the clock port is **700ps.** (HINT: source latency) 3. The maximum insertion delay from the clock port to all the register clock pins is **300ps +/- 30ps.** (HINT: Treat the 300ps as network latency and the +/-30ps as clock skew) 4. The clock period can fluctuate **+/- 40ps** due to jitter. 5. Apply **50ps** of "setup margin" to the clock period. 6. The worst case rise/fall transition time of any clock pin is **120 ps.**
Register Setup Time	Assume a maximum setup time of **0.2ns** for any register in MY_DESIGN
Input Ports (sequential logic)	1. The maximum delay from ports `data1` and `data2` through the *internal input logic* `S` is **2.2ns.** 2. The latest *F3 data arrival time* at the `sel` port is **1.4ns** absolute time. (HINT: Input delay is specified as relative time – relative to the launching clock edge)
Output Ports (sequential logic)	1. The maximum delay of the *external* combo logic at port `out1` is **420ps**; `F6` has a setup time of **80ps.** 2. The maximum *internal* delay to `out2` is **810ps** 3. The `out3` port has a **400ps** setup time requirement with respect to its capturing register clock pin.
Combinational Logic	The maximum delay from `Cin1` and `Cin2` to `Cout` is **2.45ns.** (HINT: Use appropriate input and output delay constraints with respect to clock `clk`)

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Lab Instructions

Task 1. Determine the Target Library's *Time Unit*

1. Change to the `lab3` UNIX directory.
2. Using a text editor or viewer look at the `common_setup.tcl` file to answer this question:

   Question 1. What is the *target library* file name (specified using the `TARGET_LIBRARY_FILES` variable)?

   .....

3. Invoke *Design Compiler Topographical* from the `lab3` directory:

```
UNIX% dc_shell -topo | tee -i lab3.log
```

4. Normally the *target* and *link libraries* are loaded into *Design Compiler* memory when a design is read in (with `read_verilog`, `read_vhdl`, `read_ddc`, or `analyze/elaborate`). In this specific instance we want to load the target library in DC memory so we can determine the unit of time, without loading a design. This can be done as follows:

```
dc_shell-topo> read_db <target_library_FILE>
```

   Note: You can ignore the warning about "`Overwriting design file .../sc_max.db`"

5. Determine the *library name* associated with this library file:

```
dc_shell-topo> list_libs
```

   Question 2. What is the target *library name*?

   .....

6. Generate a library report file for the above library:

```
redirect -file lib.rpt {report_lib <library_NAME>}
```


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7. Exit *Design Compiler*:

```
dc_shell-topo> exit
```

8. Use a text editor or viewer to look at the top portion of the `lib.rpt` file, and answer the following question:

   Question 3. What is the "`Time Unit`" of the target library?

   .....

9. Exit the text editor or viewer.

Task 2. Create a Timing Constraints File

1. In the `scripts` directory use a text editor to create a new file called `MY_DESIGN.con`.

   Question 4. What is the recommended first command for any constraint file?

   .....

2. Using the **Design Specification** and **Design Schematic** on the previous pages, as well as the appropriate time unit, enter the required constraints in `MY_DESIGN.con`.

   Note: You are encouraged to use the ***Job Aid*** as needed. You can also use DC's `help` and `man` commands as needed. If you are stuck you can refer to the solution file: `.solutions/MY_DESIGN.con`.

   Note: To use DC's `help` and `man` you will need to invoke the DC shell. It is also possible to access the *Design Compiler* "man" pages without invoking DC: The recommended approach is to create a separate UNIX alias, ***dcman***, for example:

```
UNIX% alias dcman "/usr/bin/man -M $SYNOPSYS/doc/syn/man"
UNIX% dcman create_clock
```

   Note: Exit the man page by typing '`q`', the lower-case [Q] key.

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3. After completing the constraints file, check your constraint syntax, and correct as necessary:

```
UNIX% dcprocheck scripts/MY_DESIGN.con
```

Note: `dcprocheck` is a syntax checking utility that is included with the *Design Compiler* executable. It is available if you are able to launch *Design Compiler* – no additional user setup is required. In a later lab, if used to check a "run script", you may ignore the warning "`read_verilog` is unknown" – `dcprocheck` prefers the command `read_file -f verilog`.

Task 3. Apply Constraints and Validate

1. Before invoking *Design Compiler*, type `ls` in the `lab3` directory and notice the sub-directory called `MY_DESIGN_LIB`: This is the *Milkyway design library* which was created when you first invoked Design Compiler in Task 1.
2. Now invoke the *DC shell* from the `lab3` directory.

   Question 5. Why do you NOT get an error message about trying to create a MW design library that already exists? (HINT: Look at the `dc_setup.tcl` file)

   .....

   .....

3. *Read*, *link* and *check* the design `rtl/MY_DESIGN.v`.
4. Apply the constraints file and make any corrections as needed:

```
source scripts/MY_DESIGN.con
```

5. Check that there are no missing or conflicting key constraints – correct as needed:

```
check_timing
```


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Note: The message "`Warning: there are 21 input ports that only have partial input delay specified. (TIM-212)`" appears if your `set_input_delay` commands have a "`-max`" option without a corresponding "`-min`" option. Since we are only constraining for setup timing, the warning is expected, and can be ignored.

6. Verify the clock and port constraints – correct as needed:

```
report_clock
report_clock -skew
report_port -verbose
```

7. Write out the applied constraints, in expanded form, to a file for further checking:

```
write_script -out scripts/MY_DESIGN.wscr
```

8. Ensure that your constraints are complete and correct by perfoming a UNIX "diff" between your write-script file, and the provided solution file – correct as needed:

```
UNIX% tkdiff scripts/MY_DESIGN.wscr
.solutions/MY_DESIGN.wscr
  OR
UNIX% diff scripts/MY_DESIGN.wscr
.solutions/MY_DESIGN.wscr
```

9. If the above "diff" command uncovers differences which you do not understand, take a look at `.solutions/MY_DESIGN.con`: This file contains comments explaining how each constraint was determined.
10. Save the *unmapped* design and exit *Design Compiler*:

```
write -format ddc -hier -out unmapped/MY_DESIGN.ddc
exit
```


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You have completed the "Timing Constraints" lab of the Design Compiler Workshop.

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Answers / Solutions

Question 1. What is the *target library* file name? (specified using the `TARGET_LIBRARY_FILES` variable)?

`sc_max.db`

Question 2. What is the target *library name*?

`cb13fs120_tsmc_max`

Question 3. What is the "Time Unit" of the target library?

1ns

Question 4. What is the recommended first command for any constraint file?

```
reset_design
```

Note: If there are multiple constraint files which are sourced sequentially, then `reset_design` should be used only once, at the top of the constraint file which is first applied.

Question 5. Why do you NOT get an error message about trying to create a MW design library that already exists? (HINT: Look at the `dc_setup.tcl` file)

In the `dc_setup.tcl` file there is an "`if`" statement which checks for the existence of a MW design library, and skips the `create_mw_lib` command if it already exists. If the script were to execute the `create_mw_lib` command and the library already existed, you would get the following message:
"`Error: Library 'MY_DESIGN_LIB' already exists. (MWUI-004)`".

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4

Environmental Attributes

Learning Objectives

After completing this lab you should be able to:

- Define *Design Compiler* environmental attributes based on a provided design schematic and specification
- Apply the attributes to a design
- Verify the applied attributes

Lab Duration: 45 minutes

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Lab Instructions

Perform the steps below. Refer to the *Lab 3* step-by-step instructions as needed.

From the `lab3/lib.rpt` file write down the library *Capacitive Load Unit*: _____

↓

Copy and rename the `lab3/scripts/MY_DESIGN.con` constraints file to `lab4/scripts/MY_DESIGN.con`

↓

Add the environmental attributes from the ***Specification*** on page 4-4 to `MY_DESIGN.con`

↓

Check and correct the syntax:

```
help; man;
UNIX% dcprocheck
```

↓

Invoke Design Compiler from the `lab4` directory

↓

Read `rtl/MY_DESIGN.v`

↓

Apply and verify the constraints and attributes; Correct as needed:

```
source; report_port -v
report_design
```

↓

Write out (`write_script`) a constraints file and "*diff*" against `.solutions/MY_DESIGN.wscr`

↓

Look at `.solutions/MY_DESIGN.con` and correct as needed

↓

Save the design in the `unmapped` directory and exit *Design Compiler*

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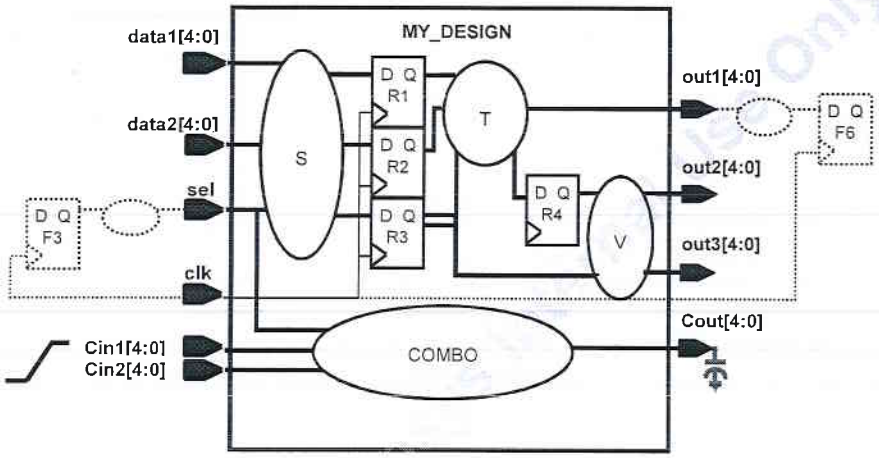
Design Schematic



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Design Specification

Hint: Use the `lib.rpt` file in `lab3` to obtain some of the information needed to apply these specs.

Input Ports (drivers)	1. Specify a drive on all inputs, except `clk` and `Cin*`, using the buffer `bufbd1` in the library 2. The `Cin*` ports are chip-level inputs and have a **120ps** maximum input transition.
Output Ports (loads)	1. All outputs, except `Cout`, drive a maximum load equivalent to **2** times the capacitance of the "`I`" pin of the cell `bufbd7` (see **Note** below). 2. The `Cout` port drives a maximum load of **25 fF**.

Note: You may get the syntax warning below if you run `dcprocheck`. The checker is overly "picky" compared to *Design Compiler* in this instance – you can ignore the warning:

```
... use curly braces to avoid double substitution
        expr 2 * [load_of lib/cell/pin]
                 ^
```

To avoid the warning enter:

```
expr {2 * [load_of lib/cell/pin]}
```

You have completed the "Environmental Attributes" lab of the Design Compiler Workshop.

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5 Synthesis Techniques

Learning Objectives

After completing this lab, you should be able to:

- Apply the recommended synthesis techniques to meet the required constraints
- Verify applied directives and variables before compile
- Analyze the gate level netlist:
  - a) To ensure that all constraints have been met
  - b) To observe the results of the various optimization techniques invoked
- Use the *Job Aid* to find all the necessary commands to accomplish the above steps
- Use the *LayoutWindow* to verify the applied physical (floorplan) constraints and view the cell placement
- Perform formal verification on the synthesized netlist versus the RTL design using Formality

Lab Duration: 75 minutes

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Lab Overview

The objective of this lab is to compile a design called STOTO using the information listed in the **Synthesis Specification** table. You will accomplish this by following the step-by-step **Lab Instructions**, which will guide you in performing the following tasks:

- Use the DC shell interactively to read in and constrain the design
- Interactively apply and execute the appropriate compile flow techniques and commands, to achieve the stated **Design Specification**
- Verify the applied compile directive commands and variables before each compile or optimization
- Analyze the results after each compile or optimization to determine what step, if any, to perform next
- While performing all the above steps interactively you will also create a "run script" so that your steps can be easily corrected and re-applied as necessary
- Use the *LayoutWindow* to verify that the applied physical (floorplan) constraints were honored during compile, and to view the cell placement
- Optionally, perform equivalence checking of the compiled netlist to RTL using *Formality*

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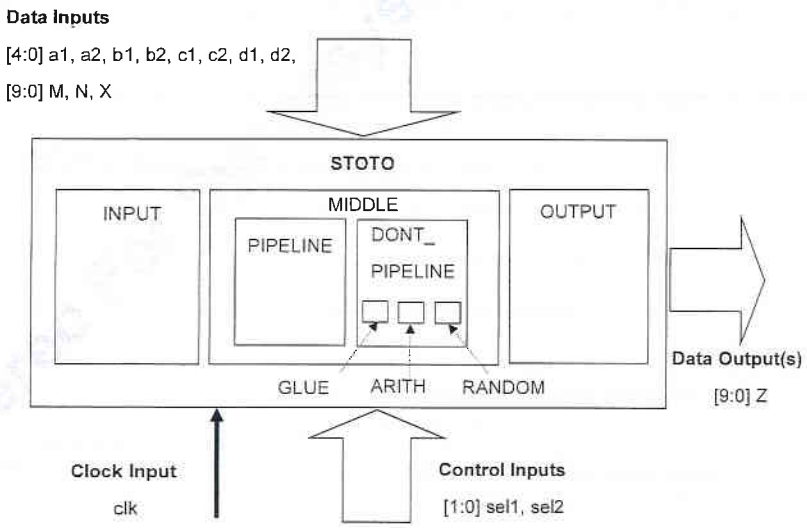
Design Schematic

You are provided with the RTL code of the design (`rtl/STOTO.v`) and the design constaints (`scripts/STOTO.con`).

RTL design hierarchy:

Design Name	Cell Name
STOTO	
INPUT	(I_IN)
MIDDLE	(I_MIDDLE)
PIPELINE	(I_PIPELINE)
DONT_PIPELINE	(I_DONT_PIPELINE)
GLUE	(I_GLUE)
ARITH	(I_ARITH)
RANDOM	(I_RANDOM)
OUTPUT	(I_OUT)



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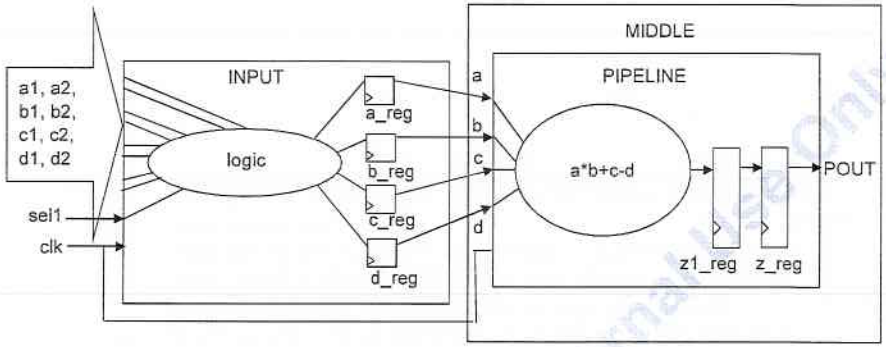

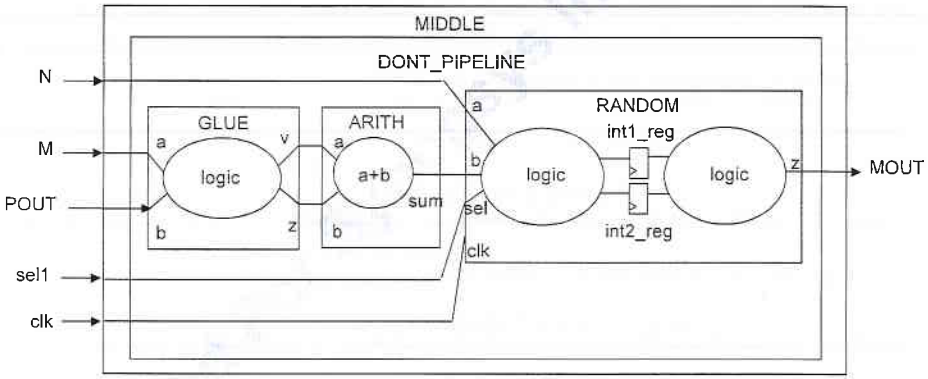

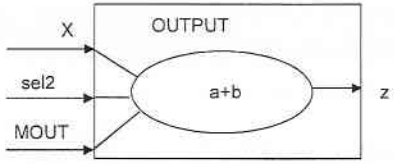

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Synthesis Specification

Available Resources	1. You will use a provided script, `num_core.sh`, to determine how many cpu cores to use 2. All Design Compiler features and related licenses are available; Two licenses are available for each feature
Design and Constraints Files	1. RTL code location: **`rtl/STOTO.v`** 2. Design Name: **`STOTO`** 3. Constraints file location: **`scripts/STOTO.con`** 4. These files may not be modified
Floorplan	Final floorplan definition: 1. The core size is 150 um X 100 um 2. STOTO has no instantiated hard macros 3. Standard cells can be placed anywhere inside the core area except for a 20 um X 20 um area in the upper-left corner, which is reserved for a different design block
Design Specification	1. The I/O constraints are estimates and have been conservatively constrained 2. The final compiled design must meet setup timing on all internal register-to-register paths 3. The **`INPUT`** block hierarchy should be preserved to facilitate post-synthesis verification 4. The **`PIPELINE`** block contains a "pure" pipelined design 5. The output (**`POUT`**) of **`PIPELINE`** must remain registered 6. The positions of registers in the **`DONT_PIPELINE`** block are fixed and cannot be modified 7. DRC violations should be fixed unless they prevent setup timing constraints from being met 8. Registers may be replicated to improve timing 9. Would like to achieve good timing correlation to post-placement timing *IC Compiler* 10. The logic positions of registers may be modified unless expressly prohibited by above specs 11. Scan insertion will be performed by the Test group after the design has met these specifications

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Task 1. Synthesis of RTL design containing pipelines

1. Using the provided `num_cores.sh` script, determine how many cores are available on your machine:

   Note: You will have to edit the file if running on *SunOS/Solaris*.

```
UNIX% ./num_cores.sh
```

   Question 1. How many cores are available on your machine?

   .....

   Note: You are encouraged to check your answers against the ***Answers/Solutions*** section at the end of the lab.

2. Look at the **Available Resources** section of the **Synthesis Specification** table to answer the following question:

   Question 2. How many cores will you be able to use to compile your design?

   .....

   Note: You will use the above information in an upcoming step to enable multi-core optimization prior to compiling the design

3. Invoke *DC shell* in the Topographical mode from the **`lab5`** directory.
4. Create a new file called **`dc.tcl`** in the **`scripts`** directory. For each of the following steps, whenever you apply a command interactively in the DC-shell environment, copy and paste the command into the **`dc.tcl`** file.
5. In your *Job Aid*, locate the **Run Script** section and the **Compile Flow** section. Refer to these sections while performing each of the following steps.
6. Before reading in the design, specify an *SVF* file name for *Formality* so that all the retiming (and other *Ultra*) transformations can be captured into a file named **`STOTO.svf`**:

```
set_svf STOTO.svf
```


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7. Refer to the **Design and Constraints Files** section of the **Synthesis Specification** table to do the following:
In the DC-shell environment, interactively read, link and check the design `STOTO`, then source and check the timing constraints.

Note: The information messages about "`preferred routing direction`" are normal and can be ignored.

Note: The `check_timing` warning message `TIM-212` about "`74 input ports that only have partial input delay specified`" can be ignored. It refers to the fact that we have only specified a `-max` delay option without its `-min` counterpart, which is OK, since we do not worry about min-delay optimization during synthesis.

Note: Remember to also "copy and paste" the commands into your run script file `dc.tcl`!

8. Refer to the **Floorplan** section of the **Synthesis Specification** table. Source the following file to apply physical constraints, which define the final floorplan:

Note: Since the floorplan definition is final, the synthesis to be performed in this lab is considered to be the "2nd pass synthesis".

```
source STOTO.pcon
```

9. **Interactively apply** the appropriate commands, up until but NOT including the first *compile*, which address **Design Specification** #1 through #7.

Remember to copy and paste into your run script.

Note: Refer to your *Job Aid* sections called **Run Script** and **Compile Flow**.

Note: If you are **really stuck** you can refer to the *run scripts* in the `.solutions` directory:
`dc.tcl` is the "plain" script which contains only the required run script commands – no "checking" and no explanations;
`dc_w_check_expl.tcl` contains additional "checking" commands to verify each applied directive, variable or attribute, as well as comments explaining each step.

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10. Enter the following commands to verify that the appropriate directives and attributes have been correctly applied prior to *compile*:

```
report_path_group
get_attribute [get_designs "INPUT"] ungroup
get_attribute [get_designs "PIPELINE"] \
        optimize_registers
get_attribute [get_cells \
        I_MIDDLE/I_PIPELINE/z_reg*] dont_retime
get_attribute [get_cells \
        I_MIDDLE/I_DONT_PIPELINE] dont_retime
get_attribute [get_designs "STOTO"] cost_priority
```

Question 3. Are you seeing the expected results for each check?

.....

.....

.....

.....

.....

Note: Check your results against the answers in the *Answers/Solutions* section at the end of this lab. If you do not see the expected results modify your applied commands until you do.

11. Save your design as `unmapped/STOTO.ddc`.

12. Apply the appropriate command to enable **multi-core optimization**, based on your answer to Question #2 (assuming more than 1 core is available).

13. Look at **Design Specification** #8 - #11, and answer the following question:

Question 4. What `compile_ultra` options will you apply?

.....

14. **Compile** the design using the options identified in the previous question.

15. While you are waiting for the compile to finish, use the design schematic on the earlier pages to answer the following two questions:

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Question 5. Is the design well partitioned for synthesis? Explain.

.....

.....

.....

Question 6. Which sub-designs would need to be ungrouped to obtain ideal partitioning for synthesis?

.....

.....

16. After the *compile* is completed, use the compile log "`Information`" messages to answer the following questions:

Question 7. What information message code confirms that multi-core optimization (if specified) was honored? (HINT: Look near the top of the log)

.....

.....

Question 8. What is an indication that the design is being retimed?

.....

.....

Question 9. What design hierarchies are being auto-ungrouped during `compile_ultra`? Do these match your requirements?

.....

.....

.....

17. You can also verify the design's remaining hierarchy with:

```
report_hierarchy -noleaf
```


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18. Generate a constraint report (`rc`) to the screen as well as to a file:

```
redirect -tee -file rc_compile_ultra.rpt {rc}
```

Question 10. Are there any register-register setup timing violations? If so, describe them.

.....

.....

.....

Question 11. Should you be concerned about these violations? Explain.

.....

.....

.....

.....

.....

.....

19. Record the setup timing "WNS" or "worst negative slack":

`max_delay/setup timing` WNS: _____

20. Check to see if there are any replicated registers. Recall that replicated registers, by default, are re-named, ending with "`_rep#`" :

```
# Verify the register replication naming style
printvar register_replication_naming_style
# List replicated registers, if any:
filter_collection [all_registers] \
                  "full_name =~ *_rep*"
```


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Question 12. Are there any replicated registers?

.....

21. Generate a timing report (`rt`):

```
redirect -tee -file rt_compile_ultra.rpt {rt}
```

22. Look at the "**`Startpoint`**" of the **`OUTPUTS`** group. Notice that the startpoint "**`I_MIDDLE/I_DONT_PIPELINE/I_RANDOM`**..." contains the instance names of sub-designs that were auto-ungrouped!

When ungrouping, *Design Compiler* keeps the original hierarchical path name to a child cell and converts it into a non-hierarchical cell name. This register start-point is actually a leaf cell in the top-level design `STOTO`!! The "slash" no longer denotes a hierarchy separator but is just a character that is part of the cell's new, longer name. This makes it easy to "trace" ungrouped cells to know where they originally came from.

23. Save your design as **`mapped/STOTO.ddc`** for IC Compiler.

24. Stop recording design changes in the *SVF* file for *Formality*:

```
set_svf -off
```

Next we are going to determine if registers were affected by adaptive retiming or by register retiming.

The cell name of a pipeline register that has been moved or repositioned by *Register Retiming* is re-named: **`clockname_r_REG#_S#`**.

25. Enter the following command. If it returns any cells, this confirms that these registers were moved by *register retiming*:

```
get_cells -hier *r_REG*_S*
```

Look at the returned cells to answer the following questions.

Question 13. In what sub-design are these retimed registers?

.....

Question 14. How can you verify that the <u>output</u> registers in **`PIPELINE`**, called **`z_reg*`**, were <u>not</u> moved?

.....

.....

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Non-pipeline registers moved by *Adaptive Retiming* are named **`R_##`**.

26. Enter the following command. If it returns any cells, this confirms that these registers were moved by *adaptive retiming*:

```
get_cells -hier R_*
```

Look at the returned cells to answer the following questions.

Question 15. Were registers in the **`INPUT`** sub-design affected by adaptive retiming, and if so, is this due to the `-retime` option?

.....

.....

Question 16. Were registers in the **`PIPELINE`** design affected by adaptive retiming, and if so, is this due to the `-retime` option?

.....

.....

Question 17. Were ALL registers in the **`INPUT`** sub-design affected by adaptive retiming?

.....

.....

27. Refer to the bottom portion of the "**Compile Flow**" section of the *Job Aid* to answer the following two questions:

Question 18. What are the next suggested steps if there are still violations after the initial compile?

.....

.....

.....

.....

.....

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Question 19. Based on the **Design Specification**, does it make sense to perform these steps?

.....

.....

.....

.....

.....

.....

28. We will next see if we can fix the remaining DRC violation, by reverting back to the "default" cost priority of DRC over timing, and performing an incremental compile:

```
set_cost_priority -default
compile_ultra -spg -retime -scan -incremental
```

29. Generate a constraints report:

```
redirect -tee -file rc_incr_compile.rpt {rc}
```

Question 20. Has the DRC violation been fixed?

.....

30. Record the setup timing "WNS" or "worst negative slack":

`max_delay/setup timing` WNS: _____

Question 21. How does this WNS compare to the WNS after the initial compile (see step 19)?

.....

.....

Next you will bring up the layout window (discussed in more detail in a later unit) to view the floorplan and cell placement.

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31. Open a *LayoutWindow* from the *DesignVision* GUI as follows:

```
dc_shell-topo> start_gui
Design Vision GUI: Window → New Layout Window
```

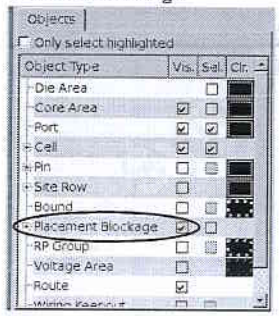
You should see a rectangular shape, which represents the core placement area, as well as the input/output pin locations along the boundary of the core. The placed cells are not visible, by default.

32. Place your cursor on the upper-right corner of the core boundary and look at the (X,Y) coordinates in the bottom right banner of the *LayoutWindow*.

Question 22. Does the size of the placement area match the physical constraint applied in `./scripts/STOTO.pcon`?

.....

33. From the "Objects" form on the left side of the *LayoutWindow*, turn on the visibility of placement blockages by checking the visibility box next to **Placement Blockage**:



The placement blockage is highlighted in the upper-left corner of the placement area.

Note: You can toggle between showing and hiding the "Objects" form by using the *View* → *Toolbars* → *View Settings* pull down (or) the F8 key.

Question 23. Does the location of the placement blockage match the physical constraint applied in `./scripts/STOTO.pcon`?

.....

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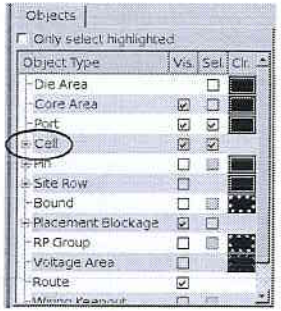
34. Expand the **Cell** objects by clicking on the "+" icon:



35. Check the "**Vis.**" (Visible) and "**Sel.**" (Selectable) boxes next to "**Standard**":

The standard cells should now be visible. You can use the "Z" and "F" keys to zoom in and fit the layout view.

You may notice that some standard cells overlap, and their placement looks "un-organized". Standard cells must eventually be placed in "legal" placement locations (no overlapping) inside "placement rows" (usually running horizontally), so their top and bottom edges align perfectly. This is not the case here because *DC-Topo* performs only "coarse placement", for quicker placement – it does not perform placement "legalization". Coarse placement is good enough for purposes of estimating the interconnect or net parasitic R/C's, which is the main goal of *DC-Topographical* synthesis.

36. Exit Design Compiler.

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Task 2. OPTIONAL: Formal Verification

In this task, you will invoke *Formality* (Synopsys' Formal Verification tool) and verify that your RTL (the "Reference") and the optimized netlist (the "Implementation") are functionally equivalent.

1. Open the `./scripts/fm.tcl` file to get an understanding of the *Formality* flow.

   Question 24. Which file contains *SVF* information written out by DC?

   .....

   Question 25. Which files are the "Reference" and "Implementation" design files that are being verified for functional equivalence?

   .....

2. From the `lab5` directory invoke the *Formality* shell:

```
UNIX% fm_shell
```

3. Execute the `fm.tcl` script:

```
fm_shell (setup)> source -echo ./scripts/fm.tcl
```

   Question 26. Does verification succeed or fail?

   .....

   Note: If the verification failed, check if the *svf* file has been read in successfully.

A successful verification means that the gate-level netlist generated by synthesis is logically equivalent to the original RTL, which is the purpose of equivalency checking. Your work is done!

If you want to get an idea of the changes or transformations that DC performed and were captured in the *SVF* file, look at the last table in the log (generated by the command `report_guidance -summary`). For example, from the listed guidance commands "`retiming`" and "`ungroup`" you know that registers have been retimed, and sub-designs have been ungrouped. These RTL-to-netlist changes are passed on to *Formality*, in the form of "guidance commands", to help *Formality* in verifying logical equivalence. The guidance commands are applied to the "reference" input (the

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RTL design, in this case) during the "matching" phase. During the "verification" phase *Formality* compares the functionality between the modified reference input (modified by "accepted" guidance commands) and the "implementation" input (the gate-level netlist, in this case). If the reference input and the implementation input are logically equivalent, the verification passes.

Note that an "accepted" guidance command means that *Formality* was able to verify that a specific transformation performed by DC was logically valid. Your netlist can still fail logical equivalency verification with 100% accepted guidance commands, and conversely, verification can pass even if there are some rejected guidance commands.

If you run *Formality* without providing the *SVF* file, the tool may not be able to verify equivalency.

For more information on *Formality* please refer to the *Formality User's Guide*, available on-line through *SolvNet*.

4. Exit *Formality*.

You have now seen how *Ultra* optimization techniques such as register retiming and adaptive retiming, auto-ungrouping, and datapath optimization can be effectively used to improve timing in a timing-critical design. With the help of the provided *Job Aid* you should now be able to apply the recommended synthesis flow and techniques to any real-world design!

You have completed the Synthesis Techniques lab of the Design Compiler Workshop.

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Answers / Solutions

Question 1. How many cores are available on your machine?

Your answer may vary. In the following example, 8 cores are available:

```
Linux ......
processor       : 0
processor       : 1
processor       : 2
....
processor       : 7
```

Question 2. How many cores will you be able to use to compile your design?

Since you have 2 licenses available and each license enables you to use 4 cores, you can specify up to 8 cores during compile. If your answer to Question 1 was 8 or more, you will be able to specify 8 cores. If your answer was less than 8 cores, you can use the exact number of cores that are available.

Question 3. Are you seeing the expected results for each check?

`report_path_group` should confirm that there are 4 path groups, in addition to the `default` group: an input, output, combinational and `clk` path group. The `clk` group should have a *weight* of `5.00` and a *critical range* of `0.21`, while the other groups should have `1.00` and `0.00`, respectively. (Spec #1 and #2)

`get_attribute [get_designs "INPUT"] ungroup` should return "**`false`**" as the value of the ungroup attribute on each of the two designs. (Spec #3)

`get_attribute [get_designs "PIPELINE"] \ optimize_registers` should return "**`true`**" as a result of applying `set_optimize_registers` on the "`PIPELINE`" block. (Spec # 4)

`get_attribute \ [get_cells I_MIDDLE/I_PIPELINE/z_reg*] \ dont_retime` should return "**`true true true true true true true true true true`**" as a result of

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using the `set_dont_retime` command on these registers. (Spec #5)

`get_attribute \ [get_cells I_MIDDLE/I_DONT_PIPELINE] \ dont_retime` should return "**true**" as a result of using the `set_dont_retime` command on the "`DONT_PIPELINE`" block. (Spec # 6)

`get_attribute [get_designs "STOTO"] \ cost_priority` should return "**max_delay**" as a result of applying `set_cost_priority`.(Spec #7)

Question 4. What `compile_ultra` options will you apply?

```
compile_ultra -spg -retime -scan
```

From #8, #9 and #10, you should use **-spg** option: This enables register replication, netlist topology congestion optimization, and includes coarse placement information in the *ddc* netlist, which can be used as a start-point by *IC Compiler*'s placement, to achieve good correlation between synthesis and post-placement timing.

From #11 "The logic positions of registers may be modified unless expressly prohibited by above specs", you should use **-retime**.

From #12 "Scan insertion will be performed by the Test group after the design has met these specifications", you should use the **-scan** option.

Question 5. Is the design well partitioned for synthesis? Explain.

No! Logic optimization will be restricted at the interfaces between the following sub-designs, due to hierarchical partitioning:

`GLUE` → `ARITH`
`ARITH` → `RANDOM`
`RANDOM` → `OUTPUT`

Question 6. Which sub-designs would need to be ungrouped to obtain ideal partitioning for synthesis?

`MIDDLE`, `DONT_PIPELINE`, `GLUE`, `ARITH`, `RANDOM` and `OUTPUT`.

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Question 7. What information message code confirms that multi-core optimization (if specified) was honored? (HINT: Look near the top of the log)

```
Information: Running optimization using
a maximum of 8 cores. (OPT-1500)
```

Also, if you enter "`list_licenses`", you will find both licenses in use:

```
...
DC-Ultra-Features (1)
...
```

Question 8. What is an indication that the design is being retimed?

```
Information: Retiming is enabled. SVF
file must be used for formal
verification. (OPT-1210)
```

This refers to either register or adaptive retiming.

Also, after the beginning of "**Mapping Optimizations (Ultra High effort)**" there are "**Retiming**" messages indicating the register retiming steps being performed on the **PIPELINE** design.

Question 9. What design hierarchies are being auto-ungrouped during `compile_ultra`? Do these match your requirements?

OPT-776 messages should confirm that the six designs listed in *Question 6* are ungrouped before "`Pass 1 Mapping`". Per the spec, the "`INPUT`" block should not be ungrouped.

Question 10. Are there any register-register timing violations? If so, describe them.

No, there are only **max-delay** or **setup** timing violations in the **INPUTS** path group.

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Question 11. Should you be concerned about these violations? Explain.

Since, according to the **Design Specification**, the I/O constraints are conservative, and the goal is to meet reg-to-reg timing, the max-delay violations are not critical (Spec #1 and 2).
From Design Spec #7, this DRC violation is most likely a result of increasing the "cost priority" of setup timing over DRCs. The violating net is in the "`INPUT`" sub-design, whose timing paths are conservatively constrained and are therefore in their own path group called "`INPUTS`". If fixing this DRC is considered more important than the timing violations, an incremental compile can be performed with "default" cost priority, to try to eliminate this violation. We will attempt this later in the lab. It is OK to ignore if there are any DRC violations at this point.

Question 12. Are there any replicated registers?

No. Although we enabled register replication using `compile_ultra -spg`, apparently, there were no timing-critical paths which could have improved, or needed to be improved, via register replication. Other, less "costly" *Ultra* optimization techniques were able to achieve the required timing.

Question 13. In what sub-design are these retimed registers?

The cell names all begin with **`I_MIDDLE/I_PIPELINE/clk*`**, which indicates that the retimed registers are in the `PIPELINE` sub-design, or "reference". You can confirm this "cell name" versus "reference name" relationship with
`report_cell -nosplit I_MIDDLE/I_PIPELINE`

Question 14. How can you verify that the output registers in **`PIPELINE`**, called **`z_reg*`**, were not moved?

Since all the register cells end with **`_S1`** you know that only **`z1_reg`**, the first stage registers, were moved.

This can be further verified with additional checks:
`get_cells -hier *z_reg*` shows that the original register names still exist, which means that *`Register Retiming`* did not move them. You can also generate a timing report which shows that the `PIPELINE` output is registered:

```
report_timing \
     -from I_MIDDLE/I_PIPELINE/z_reg*/*
```


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Question 15. Were registers in the **`INPUT`** sub-design affected by adaptive retiming, and if so, is this due to the `-retime` option?

Yes, **`I_IN`** is the instance name of the **`INPUT`** sub-design. You can confirm this cell versus "reference" name relationship with `report_cell -nosplit I_IN`. These registers were moved because of the `-retime` option used with `compile_ultra`. We know this because of the register cell names, which follow the *adaptive retiming* naming convention of `R_#`.

Question 16. Were registers in the **`PIPELINE`** design affected by adaptive retiming, and if so, is this due to the `-retime` option?

No, there are no registers called **`R_#`** in the **`PIPELINE`** design. If there were, this would mean that they were moved by adaptive retiming as well. However, this adaptive retiming would NOT have been due to `-retime`, but due to register retiming. The final optimization phase of register retiming includes an implicit adaptive retiming phase.

Question 17. Were ALL registers in the **`INPUT`** sub-design affected by adaptive retiming?

No! Using `get_cells I_IN/*_reg*` you can confirm that some of the registers still have their original register names, which confirms that they were NOT affected by adaptive retiming.

Question 18. What are the next suggested steps if there are still violations in the register-to-register paths?

1) Apply more focus on violating critical paths, as necessary: `group_path -weight -critical`

2) Perform an incremental ultra compile
`compile_ultra -scan -retime -spg -inc`

Question 19. Based on the **Design Specification**, does it make sense to perform these steps?

It depends. We have already met the stated specification that all reg-to-reg setup timing must be met. Since the I/O constraints are estimates and have been conservatively constrained, it does not make sense to spend any more time trying to get these I/O violations to pass. However, there is a DRC violation (see Questions 10 and 11) which may be fixable, possibly at the expense of worse timing in the

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INPUT block. It may be more important to try to meet the DRC, even at the expense of INPUT timing. For this lab, we'll assume that meeting DRCs in the INPUT block is more important than meeting timing.

Question 20. Has the DRC violation been fixed?

Yes!

Question 21. How does this WNS compare to the WNS after the initial compile (see step 19)?

It has also improved. Since the reg-to-reg paths already meet timing, the incremental compile was able to focus solely on the remaining violations, and was, apparently, able to improve them somewhat.

Question 22. Does the size of the placement area match the physical constraint applied in `./scripts/STOTO.pcon`?

It should match the placement area constraint shown below.

Question 23. Does the location of the placement blockage match the physical constraint applied in `./scripts/STOTO.pcon`?

It should match the placement blockage constraint below.

```
set_placement_area -coordinate {0 0 150 100}
create_placement_blockage -name Blockage1 \
   -coordinate {0 80 20 100}
```

You can also confirm the physical constraints with:

```
report_physical_constraints
```

Question 24. Which file contains SVF information written out by DC?

`STOTO.svf`

The file name was specified using the `set_svf` command prior to synthesis, and is repeated here for *Formality*. Any optimization transformations which can affect formal verification are recorded by DC into this binary file. If the user does not specify a file name, *SVF* data is written to a file named `default.svf`. A text equivalent *SVF* file, called `svf.txt` can be found under the `formality_svf` directory after running *Formality*.

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Question 25. Which files are the "Reference" and "Implementation" design files that are being verified for functional equivalence?

The reference RTL design file is **`rtl/STOTO.v`**.
(`read_verilog -r STOTO.v`)

The implementation or gate level design file is **`mapped/stoto.retime.ddc`**.
(`read_ddc -i STOTO.ddc`)

Question 26. Does verification succeed or fail?

Near the end of the log output, under `*** Verification Results ***` you should see the message "**`Verification SUCCEEDED`**"

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6 Timing Analysis

There is NO LAB for this unit

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7

Additional Constraint Options

Learning Objectives

After completing this lab you should be able to:

- Constrain a design with the following situations:
  - Contains positive and negative edge-triggered registers
  - Inputs/outputs are being driven/captured by multiple paths
  - External input/output paths have non-default latencies
  - Input drivers fan out to additional external loads
- Compile the constrained design
- Relate entries in a timing report to their corresponding constraints

Lab Duration: 75 minutes

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Instructions

Perform the steps below.

Copy and rename the `lab4/scripts/MY_DESIGN.con` constraints file to `lab7/scripts/MY_DESIGN.con`

↓

Complete the constraints based on the ***Specification*** and ***Schematic*** on the next pages

↓

Check and correct the syntax as done in *Lab 3* and *Lab 4*

↓

Invoke Design Compiler from the `lab7` directory

↓

Read `./rtl/MY_DESIGN.v`

↓

Apply, verify and correct the constraints, as done in *Lab 3* and *Lab 4*. Compare against `.solutions/MY_DESIGN.wscr` or `.solutions/MY_DESIGN.con` as needed. Use the **"latencies included"** method for this lab.

↓

Compile the design using:

```
compile_ultra -scan \
              -retime
```

↓

Answer the **"Lab 7: QUESTIONS"** starting on page 7-5

↓

Save the design in the `mapped` directory and exit *Design Compiler*

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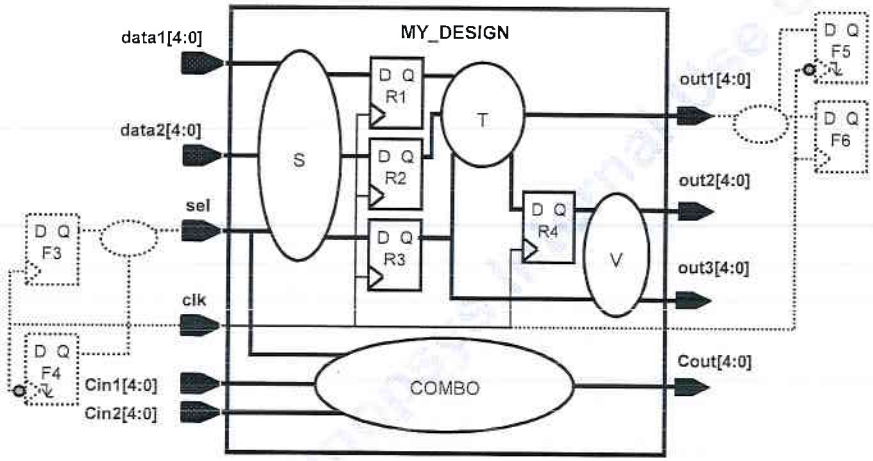
Design Schematic



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Design Specification

Clock Spec	1. Rename the clock from "`clk`" to "`my_clk`" 2. Change the duty cycle to **40 % active high**, with **zero offset**.
Input Port Delays	1. The `sel` port data is being supplied by an additional register, `F4`. The data at `sel` arrives no later than **420ps** after the negative edge-triggered launching clock edge of `F4`. 2. The clock signal at `F4` has a **600ps total (external + internal) insertion delay** from its clock source
Output Port Delays	1. The `out1` port data is captured by an additional register, `F5`. The data at `out1` must arrive no later than **260ps** before `F5`'s negative edge-triggered clock edge. 2. The clock signal at `F5` has an **internal latency of 500ps** after the external `my_clk` clock generator delay
External Load, Input Ports	Each input port (except `clk`) fans out to **2 other sub-blocks**, which each drive the equivalent of **3 `bufbd1`** (input pin `I`) buffers internally. Model this external **capacitive loading** on the inputs.

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Questions

Refer to the "*Answers/Solutions*" section at the end of this lab to verify your answers or if you need some help. Note that our answers to the following questions are based on using the "latencies included" method. If you used the "virtual clocks" method, some of your answers will differ; you may compare your constraints file against the `.solutions/MY_DESIGN_w_vclk.wscr` or `.solutions/MY_DESIGN_w_vlk.con` files as needed.

1. Generate a constraint report:

```
report_constraint -all_violators
```

Question 1. Does the design have any timing or DRC violations?

.....

2. Generate a timing report for the path to the `out1` output. Include options to show net *transition times* and *net delays* to *6 decimal places*, as well as net *fanout*:

```
report_timing -trans -input -sig 6 -nets \
              -to [get_ports out1]
```

3. Use the report to locate and fill in the following requested data. This data should match your constraints:

Question 2. What is the startpoint of the reported path?

.....

Question 3. From this report identify the following startpoint or data launch values. Include the constraint file command(s) which produce each of the report values:

Clock `my_clk`
Launch edge time _____ Rising or falling? _____

Command(s) .....

.....

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Clock network delay = ...

Command(s) .....

.....

Clock pin transition time = ...

Command(s) .....

.....

Question 4. From the report identify the following endpoint or data capture values. Include the constraint file command(s) which produce each of the report values:

Clock `my_clk`
Capture edge time _____ Rising or falling? _____

Command(s) .....

.....

Clock network delay = ...

Command(s) .....

.....

Clock uncertainty = ...

Command(s) .....

.....

Output external delay = ...

Command(s) .....

.....

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Question 5. Why is the output timing with respect to a falling clock edge?

.....

.....

.....

Question 6. Why are the report values for *uncertainty* and *output external delay* the negative of their corresponding constraint values?

.....

.....

4. Generate a timing report for the path from the `sel` input to the `Cout` output. Include options to show *transition times* and *net delays* to *6 decimal places*. Use the report to locate and fill in the requested data. This data should match your constraints:

Question 7. From the report identify the following startpoint values. Include the constraint file command(s) which produce each of the report values:

Clock `my_clk`
Launch edge time _____ Rising or falling? _____

Command(s) .....

.....

Clock network delay = ...

Command(s) .....

.....

Input external delay = ...

Command(s) .....

.....

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Transition time of `sel` port = ...

Command(s) .....

.....

Question 8. What is causing the "`Incr`" (incremental) delay on the `sel` input port?

.....

Question 9. How can you verify that the COMBO path from `Cin*` to `Cout` is constrained to the required 2.45ns spec?

.....

5. Save the design in `mapped` and exit Design Compiler.

You are now able to create, verify and identify timing and environmental constraints for any single clock design given the schematic and specification !!!

You have completed the More Constraint Considerations lab of the Design Compiler Workshop.

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Answers / Solutions – Using the "Latencies Included" method

Question 1. Does the design have any timing or DRC violations?

From `report_constraint -all` you should see that there are no timing or design rule constraints violations. If you have any violations [other than any Minimum Pulse Width violations that you can ignore for our lab]:

1) Check and correct your constraints (compare against `.solutions/MY_DESIGN.con`)
2) Remove the design from memory
3) Re-apply the constraints
4) Re-compile the design
5) Generate another constraint report

Question 2. What is the startpoint of the reported path?

The startpoint is the clock pin of a register:
`R#_reg[#]/CP`

Question 3. From the report identify the following startpoint values. Include the constraint file command(s) which produce each of the report values:

Clock **`my_clk`**
Launch edge time: **0.0ns, Rising**

```
create_clock -period 3.0              \
     -name my_clk -waveform {0 1.2} \
      [get_ports clk]
```

Clock network delay = **1.0ns**

```
set_clock_latency -source -max 0.7  \
          [get_clocks my_clk]
set_clock_latency -max 0.3          \
          [get_clocks my_clk]
```

Clock pin transition time = **0.12ns**

```
set_clock_transition -max 0.12      \
          [get_clocks my_clk]
```


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Question 4. From the report identify the following endpoint values. Include the constraint file command(s) which produce each of the report values:

Clock **`my_clk`**
Capture edge time: **1.2ns, Falling**

```
create_clock -period 3.0              \
     -name my_clk -waveform {0 1.2} \
      [get_ports clk]
set_output_delay -max -0.94           \
     -clock my_clk -add_delay         \
     -clock_fall                      \
     -network_latency_included        \
     -source_latency_included         \
      [get_ports out1]
```

Clock network delay = **0.0 ns**

Note: Your results will be different if you did not use the `-network_latency_included` and/or the `-source_latency_included` options as shown here.

```
set_clock_latency -source -max 0.7  \
      [get_clocks my_clk]
set_output_delay -max -0.94         \
     -clock my_clk -add_delay       \
     -clock_fall                    \
     -network_latency_included      \
     -source_latency_included       \
      [get_ports out1]
```

Clock uncertainty = **-0.15ns**

```
set_clock_uncertainty -setup 0.15 \
      [get_clocks my_clk]
```

Output external delay = **0.94ns**

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```
set_output_delay -max -0.94        \
      -clock my_clk -add_delay     \
      -clock_fall                  \
      -network_latency_included    \
      -source_latency_included     \
       [get_ports out1]
```

Question 5. Why is the output timing with respect to a falling clock edge?

The `out1` port is constrained by two registers – a rising (`F6`) and a falling (`F5`) edge-triggered one. DC determined that the timing constraint to the falling edge-triggered register is the more restrictive of the two.

Question 6. Why are the report values for *uncertainty* and *output external delay* the negative of their corresponding constraint values?

The negated values simply mean that the constraint numbers are being subtracted from the "*data required time*".

Question 7. From the report identify the following startpoint values. Include the constraint file command(s) which produce each of the report values:

Clock **`my_clk`**
Launch edge time: **1.2ns, Falling**

```
create_clock -period 3.0              \
      -name my_clk -waveform {0 1.2} \
       [get_ports clk]

set_input_delay -max 1.02             \
      -clock my_clk -add_delay        \
      -clock_fall                     \
      -network_latency_included       \
      -source_latency_included        \
       [get_ports sel]
```

Clock network delay = **0.0ns**

Note: Your results will be different if you did not use the `-network_latency_included` and `-source_latency_included` options as shown here.

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```
set_input_delay -max 1.02          \
   -clock my_clk -add_delay        \
   -clock_fall                     \
   -network_latency_included       \
   -source_latency_included        \
          [get_ports sel]
```

Input external delay = **1.02ns**

```
set_input_delay -max 1.02          \
   -clock my_clk -add_delay        \
   -clock_fall                     \
   -network_latency_included       \
   -source_latency_included        \
          [get_ports sel]
```

Transition time of **`sel`** port = **~0.25** or more **ns**

```
set_driving_cell                   \
   -max -lib_cell bufbd1           \
   -library cb13fs120_tsmc_max     \
   [remove_from_collection         \
   [all_inputs]                    \
   [get_ports "clk Cin*"]]
```

Question 8. What is causing the "`Incr`" (incremental) delay on the **`sel`** input port?

This represents the additional time for the input signal, with the above transition time, to reach the "switching point". It is not due to net delay, which is reported separately at the input pin of the first gate.

Question 9. How can you verify that the COMBO path from **`Cin*`** to **`Cout`** is constrained to the required 2.45ns spec?

```
report_timing                      \
      -from [get_ports Cin*]       \
      -to [get_ports Cout]
```

Subtract the data arrival time at the `Cin` input port (`1.30 ns` in the "`Path`" column) from the "`data required time`" (`3.75 ns`), to get **2.45ns**.

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8 Multiple Clocks and Timing Exceptions

Learning Objectives

The goal of this lab is to give you a better understanding of how static timing analysis works and how timing exceptions are properly applied.

After completing this lab, you should be able to:

- Constrain and analyze complex multi-clock designs with mutually exclusive paths and multicycle paths.

Lab Duration: 60 minutes

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Background

The design you will be working with, called "EXCEPTIONS", has already been constrained and compiled, and saved as a "mapped" *ddc* file. The design contains parallel paths shown in the figure below ("resets" not shown). The inputs `adr_i` and `coeff`, and the output `dout` are each connected to both a combinatorial path as well as a sequential path.

The combinatorial and sequential paths have different constraints. When you perform timing analysis, you will discover that there are initially many violations – all due to incomplete constraints. Your objective is to add appropriate constraints as dictated by the design requirements. Once the design is completely and correctly constrained, the design should meet all timing constraints without needing to be re-compiled.

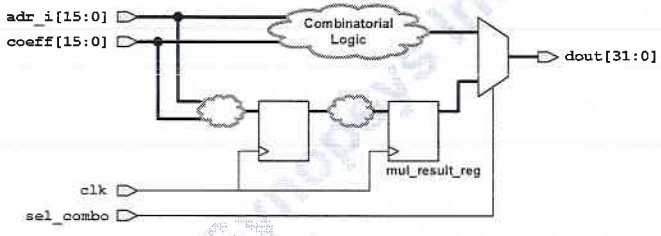

Figure 1. Design "EXCEPTIONS"

If you run into difficulties while performing the various tasks, check the Answers/Solutions section at the back of this lab.

You may also check the file `.solutions/dc.tcl` *for help.*

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Lab Instructions

Task 1. Read a Mapped Design

1. Make sure your current working directory is `lab8` and invoke `dc_shell` in *Topographical* mode .
2. Read the compiled design "`EXCEPTIONS`" into memory:

```
read_ddc EXCEPTIONS.ddc
current_design EXCEPTIONS
link
```

3. Use the "`view`" procedure to generate and analyze a constraint report and a default timing report.

```
v rc
v rt
```

Note: "`v`" is an alias for the Tcl *procedure* called "`view`", which is defined in `../ref/tools/procs.tcl`. This is not a standard *Design Compiler* command. It is very useful for viewing reports in the interactive shell, so we are making it available for your use during the labs.

Question 1. What is the timing slack of the critical path – the *worst negative slack* (WNS)?

.....

Question 2. Describe the start and end points of this violating timing path:

.....

Question 3. What is the maximum path delay constraint for this critical path (= Clock Period – Input Delay – Output Delay)?

.....

This path is over constrained - it cannot meet a **–1 ns** maximum delay! This design is improperly constrained for demonstration purposes. *The maximum propagation delay through the combinational path should be constrained to 6 ns.* The current input and output delay constraints are appropriate for the sequential paths (from the input ports to flip-flops and from flip-flops to the output ports), but clearly not for the purely

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combinational paths. The constraints must be expanded such that the combinatorial and sequential paths are constrained separately.

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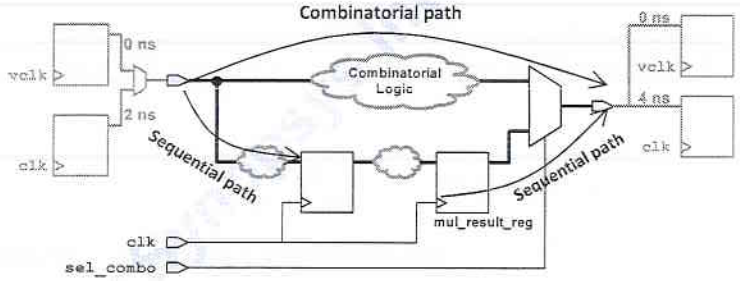
Task 2. Constrain Parallel Paths Separately

In this task you will use a *virtual clock* to enable you to constrain the combinatorial paths separately from the sequential timing paths.

The existing clock, input and output delay constraints are correct for the sequential input and output paths (see schematic below). In this task you will apply additional constraints so that the combinatorial logic paths are constrained independent of the sequential logic paths.

To accomplish this, the combinatorial paths will be constrained using a new virtual clock which you will call **vclk**. The sequential paths are already constrained using the 5ns design clock clk, and input and output delays of 2ns and 4ns, respectively. The following schematic illustrates this concept (note that the two input ports, **adr_i** and **coeff**, are collapsed into one port here to simplify the schematic):



The diagram shows that two clocks, **clk** and **vclk** are clocking the input and output data of the design. The sequential paths, to and from the internal registers, are constrained using clk, and the purely combinatorial path are to be constrained by the virtual clock vclk.

1. The combinatorial path must have a maximum delay of 6 ns. Start by creating a **6 ns** virtual clock vclk, along with input and output delays with respect to **vclk** that are set to **0 ns**.

   Remember that the input delay constraint should be applied to the ports **coeff*** and **adr_i***.

   Apply the appropriate commands based on what you learned in lecture. Use the *Job Aid* as needed.

   If you're stuck, check the Answers section.

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2. Generate a timing report for a timing path constrained by the virtual clock.

   The **-group vclk** option will generate a timing report for the path group **vclk**, which will report the timing path with the worst setup timing slack that is captured by **vclk**.

```
v rt -group vclk
```

   Question 4. Does the path violate or meet timing?

   .....

   Question 5. Describe the launch and capture clocks. Does this match your expectations?

   .....

   .....

3. The clocks **clk** and **vclk** should not interact. Apply a *logically exclusive clock group* constraint (or two *false path* constraints, if you prefer) to disable timing analysis between these clocks.

   If you're stuck, check the Answers section.

4. Repeat the previous timing report.

   Question 6. Does the path violate or meet timing?

   .....

   Question 7. Describe the launch and capture clocks and does this match your expectations?

   .....

5. The combinatorial path constraints are not complete yet! Generate the following report and answer the following questions:

```
v rt -group clk
```

   Question 8. Does this path violate or meet timing?

   .....

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Question 9. Describe the start and end points of this timing path and does this match your expectations?

.....

6. As the final step, disable timing analysis from **clk** to **clk** through the combinatorial logic using the *false path* command.

   Hint: Use the **-through** option (two are needed) to help you separate the combinatorial **clk** to **clk** paths from the sequential **clk** to **clk** paths.

7. Generate a timing report from all inputs to all outputs and confirm that the combinatorial paths are now constrained only by **vclk**, and that these paths meet timing.

Task 3. Constrain a Multicycle Path

The paths to **mul_result_reg** are supposed to be multicycle paths, which are allowed to take up to 2 clock cycles (instead of 1, the default) for setup. The hold capture checks should be done at 0ns. These paths are not correctly constrained either.

1. Execute a timing report for the clock **clk**:

```
v rt -group clk
```

Question 10. How large is this violation in comparison to the clock period?

.....

2. Apply a *multicycle path* constraint of 2 cycles for setup to all paths that end at the **D** pin of **mul_result_reg***.

3. Verify that the multicycle path exception was applied correctly:

```
v rt -to mul_result_reg*/D
```

Question 11. What is the effective clock period for this timing path (the clock capture edge minus the clock launch edge)?

.....

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4. Generate a *hold timing* report to these same end points.

```
!! -delay min
```

Question 12. What are the launch and capture clock edges and does this match the specification described at the beginning of this task?

.....

.....

5. Complete the multicycle path constraint by including a constraint for hold.

Question 13. What are the launch and capture clock edges and are they now correct for hold ?

.....

6. Generate a final "`report_constraint -all`" to make sure that all violations have been taken care of.

7. **Quit** Design Compiler.

You have completed the "Multiple Clocks and Timing Exceptions" lab of the Design Compiler Workshop.

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Answers / Solutions

Question 1. What is the timing slack of the critical path – the *worst negative slack* (WNS)?

Both the constraint and the timing report show that the **WNS is -6.78 ns**. Notice from the timing report that it is more than 100% of the 5ns clock period that is being used.

Question 2. Describe the start and end points of this violating timing path:

The start point is an input port and the end point is an output port.

Question 3. What is the maximum path delay constraint for this critical path (= Clock Period – Input Delay – Output Delay)?

-1ns (5ns – 2ns – 4ns)! This is not an achievable constraint.

Solution for task 2 step 1:

```
create_clock -name vclk -period 6
set_in_ports [get_ports "coeff* adr_i*"]
set_input_delay 0 -clock vclk -add_delay $in_ports
set_output_delay 0 -clock vclk -add_delay [all_outputs]
```

Question 4. Does the path violate or meet timing?

It violates timing.

Question 5. Describe the launch and capture clocks and does this match with your expectations?

The launch clock is **clk** and the capture clock is **vclk**. This is not correct! The combinatorial paths should be constrained only by **vclk**. The two clocks should not interact.

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Solution for task 2 step 3:

```
set_clock_group -name false_grp1 -logically_exclusive \
   -group clk -group vclk
# OR alternatively:
set_false_path -from [get_clocks clk] -to [get_clocks vclk]
set_false_path -from [get_clocks vclk] -to [get_clocks clk]
```

Question 6. Does the path violate or meet timing?

It meets timing.

Question 7. Describe the launch and capture clocks and does this match your expectations?

Both the launch and capture clocks are **vclk**, as expected.

Question 8. Does this path violate or meet timing?

It violates timing.

Question 9. Describe the start and end points of this timing path and does this match your expectations?

The start point is an input port and the end point is an output port. This should not happen! The combinatorial paths should only be constrained by **vclk** and not by **clk**.

Solution for task 2 step 6:

```
set_false_path -from [get_clocks clk] \
   -through $in_ports \
   -through [all_outputs] -to [get_clocks clk]
v rt -from $in_ports -to [all_outputs]
```


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Question 10. How large is this violation in comparison to the clock period?

The violation is **4.68 ns**, which is almost equal to the clock period of 5 ns.

Solution for task 3 step 2:

```
set_multicycle_path 2 -setup -to mul_result_reg*/D
```

Question 11. What is the effective clock period for this timing path (the clock capture edge minus the clock launch edge)?

The effective clock period is 10 ns, 2 times the actual clock period of **clk** (5 ns).

Question 12. What are the launch and capture clock edges and does this match the specification described at the beginning of this task?

The launch clock edge is 0ns and the capture clock edge is at 5 ns (one edge preceding the setup clock edge). This does not match the original specification which stated the hold capture checks should be done at zero.

Solution for task 3 step 5:

```
set_multicycle_path 1 -hold -to mul_result_reg*/D
```

Question 13. What are the launch and capture clock edges and are they now correct for hold?

The launch and capture clock edges are at zero and this is now correct.

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9

DCG: Congestion Analysis and Optimization

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10 Post Synthesis Output Data

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SYNOPSYS®

Design Compiler 2016.03-SP5 Job Aid

Invoking Design Compiler in Topographical Mode

```
unix% dc_shell -topographical              # Interactive shell
dc_shell-topo> start_gui                   # Open GUI from shell
dc_shell-topo> stop_gui                    # Close GUI, return to shell
unix% design_vision -topographical         # Open GUI directly
unix% dc_shell -topo -f dc.tcl | tee -i dc.log   # Batch mode
```

Getting Help in DC shell

```
help -verbose *clock          # Lists command options
create_clock -help            # Lists command options
man create_clock              # Displays command man page
apropos \                     # List command(s) with key word
  -symbols_only period        #  in the command or option
printvar *_library            # Lists variable and values
man target_library            # Displays variable man page
```

Settings and Checks Applied at Startup

```
# These settings can all be included in the .synopsys_dc.setup file, OR,
# they can be split up into "common_setup.tcl" and "dc_setup.tcl" files, which are
# then sourced by the "run script", called "dc.tcl", as done by the "Reference
# Methodology" generated scripts
set ADDITIONAL_SEARCH_PATH     "./libs/sc/LM ../rtl ./scripts"
set TARGET_LIBRARY_FILES       sc_max.db
set ADDL_LINK_LIBRARY_FILES    IP_max.db
set SYMBOL_LIBRARY_FILES       sc.sdb
set MW_DESIGN_LIB              MY_DESIGN_LIB
set MW_REFERENCE_LIB_DIRS      "./libs/mw_lib/sc ./libs/mw_libs/IP"
set TECH_FILE                  ./libs/tech/cb13_6m.tf
set TLUPLUS_MAX_FILE           ./libs/tlup/cb13_6m_max.tluplus
set MAP_FILE                   ./libs/tlup/cb13_6m.map

set_app_var search_path "$search_path $ADDITIONAL_SEARCH_PATH"
set_app_var target_library $TARGET_LIBRARY_FILES
set_app_var link_library "* $target_library $ADDL_LINK_LIBRARY_FILES"
set_app_var symbol_library $SYMBOL_LIBRARY_FILES
set_app_var mw_reference_library $MW_REFERENCE_LIB_DIRS
set_app_var mw_design_library $MW_DESIGN_LIB
get_app_var -list -only_changed_vars *

# Only create new Milkyway design library if it doesn't already exist
if {![file isdirectory $mw_design_library ]} {
    create_mw_lib -technology $TECH_FILE \
      -mw_reference_library $mw_reference_library $mw_design_library}
open_mw_lib  $mw_design_library
check_library
set_tlu_plus_files -max_tluplus $TLUPLUS_MAX_FILE -tech2itf_map $MAP_FILE
check_tlu_plus_files
history keep 200
set_app_var alib_library_analysis_path ./ ;# ALIB files
define_design_lib WORK -path ./work
set_svf <my_filename.svf>
set_app_var sh_enable_page_mode false
suppress_message {LINT-28 LINT-32 LINT-33 UID-401}
set_app_var alib_library_analysis_path [get_unix_variable HOME]
alias h history
alias rc "report_constraint -all_violators"
```

Additional Help

```
# Using DC "man" from UNIX prompt
unix% alias dcman "/usr/bin/man \
      -M $SYNOPSYS/doc/syn/man"
unix% dcman target_library

Solvnet articles, documentation, support:
solvnet.synopsys.com

Register for workshops:
training.synopsys.com
```

Helpful UNIX-like DC-shell commands

```
pwd; cd; ls;
history; !!; !?; !report
sh <UNIX_command>
printenv
get_unix_variable <UNIX_variable>
```

TCL Commands and Constructs

```
set PER 2.0
echo $PER            # Result: 2.0
set MARG 0.95
expr $PER * $MARG    # expr: *, /, +, -, >, <, =, <=, >=
set PCI_PORTS [get_ports A]
set PCI_PORTS [get_ports "Y??M Z*"]
echo "Effctv P = [expr $PERIOD * $MARGIN]"
# Result with soft quotes: "Effctv P = 1.9"
echo {Effctv P = [expr $PERIOD * $MARGIN]}
# Result with hard quotes:
# "Effctv P = [expr $PERIOD * $MARGIN]"

# Tcl Comment line
set COMMENT in_line ;   # Tcl inline comment

set MY_DESIGNS "A.v B.v Top.v"
foreach DESIGN $MY_DESIGNS {
  read_verilog $DESIGN}
for {set i 1} {$i < 10} {incr i} {
  read_verilog BLOCK_$i.v}
```


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Timing Constraints

```
reset_design
############# CLOCKS ##############################
create_clock -period 2 -name Main_Clk [get_ports Clk1]
create_clock -period 2.5 -waveform {0 1.5} [get_ports Clk2]
create_clock -period 3.5 -name V_Clk; # VIRTUAL clock
create_generated_clock -name  DIV2CLK -divide_by 2 -source [get_ports Clk1] \
          [get_pins I_DIV_FF/Q]
set_clock_uncertainty -setup 0.14 [get_clocks *]
set_clock_uncertainty -setup 0.21 -from [get_clocks Main_Clk] -to [get_clocks Clk2]
set_clock_latency -max 0.6 [get_clocks Main_Clk]  ; # Pre layout  ideal clock tree OR
set_propagated_clock [all_clocks] ; # Post CTS (Clock Tree Synthesis) clock tree
set_clock_latency -source -max 0.3 [get_clocks Main_Clk]
set_clock_transition -max 0.08 [get_clocks Main_Clk]
############# CLOCK TIMING EXCEPTIONS ########
set_clock_groups -logically_exclusive | -physically_exclusive | -asynchronous \
                                          -group CLKA -group CLKB
set_false_path -from [get_clocks Asynch_CLKA] -to [get_clocks Asynch_CLKB]
set_multicycle_path -setup 4 -from -from A_reg -through U_Mult/Out -to B_reg
set_multicycle_path -hold 3 -from -from A_reg -through U_Mult/Out -to B_reg
################  I/O TIMING #######################
set_input_delay -max 0.6 -clock Main_Clk \
  -source_latency_included  -network_latency_included [all_inputs]
set_input_delay -max 0.48 -clock V_Clk -clock_fall  -add_delay  [get_ports A]
set_output_delay -max 0.9 -clock Clk2 \
  -source_latency_included  -network_latency_included [get_ports "OUT2 OUT7"]
set_output_delay -max 1.1 -clock V_Clk -clock_fall  -add_delay  [get_ports OUT7]
################ ENVIRONMENT #####################
set_max_capacitance 1.2 [all_inputs]; # User-defined Design Rule Constraint
set_load -max 0.080  [all_outputs]
set_load -max [expr [load_of slow_proc/NAND2_3/A] * 4] [get_ports OUT3]
set_load -max 0.12 [all_inputs]
set_input_transition -max 0.12 [remove_from_collection [all_inputs][get_ports B]]
set_driving_cell -max -lib_cell FD1 -pin Q [get_ports B]
```

Checking and Removing Constraints

```
report_clock; report_clock -skew -attr
report_design
report_port -verbose
report_path_group
report_timing_requirements -ignored
report_auto_ungroup
report_interclock_relation
write_script -output <constraints.tcl>
check_timing
reset_path -from FF1_reg
remove_clock
remove_clock_transition
remove_clock_uncertainty
remove_input_delay
remove_output_delay
remove_driving_cell
```

Syntax Checking

Unix% dcprocheck *constr_file.con*

Misc. Reports

```
# Generate a library report file
read_db library_file.db
list_libs
redirect -file reports/lib.rpt {report_lib <libname>}
report_hierarchy [-noleaf]
# Arithmetic implementation and  resource-sharing info
report_resources
  # List area for all cells in the design
report_cell [get_cells -hier *]
```

Correlation between DC-T and ICC

```
# Apply the following ICC design planning commands:
create_bounds ...
create_rp_groups...
set_app_var placer_max_cell_density_threshold...
set_ignored_layers...

# Use the "Consistency Checker" to compare
#  settings:   Solvnet Doc Id: 026366, 2112936
```

Conservative Output Load Algorithm

Used for "load budgeting" if actual output load values are not known. Finds the largest `max_capacitance` value in the library and applies that value as a conservative output load

```
set LIB_NAME ssc_core_slow
set MAX_CAP 0
set OUTPUT_PINS [get_lib_pins $LIB_NAME/*/*   \
                                -filter "direction == 2"]
foreach_in_collection pin $OUTPUT_PINS {
  set NEW_CAP [get_attribute $pin max_capacitance]
  if {$NEW_CAP > $MAX_CAP} {
    set MAX_CAP $NEW_CAP
  }
}
set_load -max $MAX_CAP [all_outputs]
```

Physical Constraints

```
set_aspect_ratio 0.5      # Height/Width
set_utilization 0.7
set_placement_area -coordinate {0 0 600 400}
create_die_area -polygon {{0 0} {0 400} {200 400} {200 200}  {400 200} {400 0} {0 0}}
set_port_side {R} Port_N
set_port_location -coordinate {0 40} PortA
set_cell_location -coordinate {400 160} -fixed -orientation {N} RAM1
create_placement_blockage -name Blockage1 -coordinate {350 110 600 400}
create_bounds -name "b1" -coordinate {100 100 200 200} INST_1
create_site_row -coordinate {10,10} -kind CORE -space 5 -count 3 {SITE_ROW#123}
create_voltage_area -name VA1 -coordinate {100 100 200 200} INST_1
create_net_shape -type wire -net VSS -origin {0 0} -length 10 -width 2 -layer M1
create_wiring_keepouts -name "my_keep1" -layer "METAL1" -coord {12 12 100 100}
report_physical_constraints
reset_physical_constraints
```


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Run Script
(called "dc.tcl" in the RM scripts)

```
read_verilog {A.v B.v TOP.v}            or
read_sverilog {A.sv B.sv TOP.sv}        or
read_vhdl {A.vhd B.vhd TOP.vhd}         or
read_ddc MY_TOP.ddc                     or
analyze -format verilog {A.v B.v TOP.v}
elaborate MY_TOP -parameters "A_WIDTH=8, B_WIDTH=16"

current_design MY_TOP

link

if {[check_design] ==0} {

   echo "Check Design Error"

   exit                 # Exits DC if a check-design error is encountered

}                       # Continue if NO problems encountered

write_file -f ddc -hier -out unmappedd/TOP.ddc


redirect -tee -file reports/precompile.rpt {source -echo -verbose TOP.con}
redirect -append -tee -file reports/precompile.rpt {check_timing}

source <Physical_Constraints_TCL_file>  or  # Source tcl constraints, if available, or
extract_physical_constraints <DEF_file>     # Extract and apply from an existing
read_floorplan <floorplan_cmd_file>         # DEF or floorplan file


group_path -name CLK1 -critical_range <10% of CLK1 Period> -weight 5
group_path -name CLK2 -critical_range <10% of CLK2 Period> -weight 2
group_path -name INPUTS -from [all_inputs]
group_path -name OUTPUTS -to [all_outputs]
group_path -name COMBO -from [all_inputs] -to [all_outputs]

set_fix_multiple_port_nets -all -buffer_constants

*****************************************************************
"Compile Flow" commands go here – see next page
*****************************************************************

check_design
check_design -html check_design.html
sh firefox check_design.html

report_constraint -all_violators

report_timing [ -delay <max | min> ]
              [ -to <pin_port_clock_list> ]
              [ -from <pin_port_clock_list> ]
              [ -through <pin_port_list> ]
              [ -group]
              [ -input_pins ]
              [ -max_paths <path_count> ]
              [ -nworst <paths_per_endpoint_count >]
              [ -nets ]; [ -capacitance ]
              [ -significant_digits <number>]
report_qor
report_area

set_app_var verilogout_no_tri true

change_names -rule verilog -hier ;# See below to avoid bit blasting ports


# Avoiding bit-blasted ports in SystemVerilog and VHDL records
define_name_rules verilog -preserve_struct_ports
change_names -hierarchy -rules verilog

write_file -f verilog -hier -out mapped/TOP.v

write_file -f ddc -hier -out mapped/TOP.ddc

write_sdc TOP.sdc

write_scan_def -out TOP_scan.def

write_icc2_files -output TOP_icc2
exit
```

Object Retrieval and Manipulation
(Collection Commands)

```
get_ports
get_pins
get_designs
get_cells
get_nets
get_clocks
get_nets -of_objects [get_pins FF1_reg/Q]

get_libs <lib_name>
get_lib_cells <lib_name/cell_names>
get_lib_pins <lib_name/cell_name/pin_names>

all_inputs
all_outputs
all_clocks
all_registers
all_connected
all_fanin
all_fanout
all_ideal_nets


set PCI_PORTS [get_ports pci_*]

echo $PCI_PORTS                 # → _sel184

query_objects $PCI_PORTS        # → {pci_1 pci_2 ...}

get_object_name $PCI_PORTS      # → pci_1 pci_2 ...

sizeof_collection $PCI_PORTS    # → 37


set PCI_PORTS [add_to_collection $PCI_PORTS \
                 [get_ports CTRL*]]

set ALL_INP_EXC_CLK [remove_from_collection \
                 [all_inputs] [get_ports CLK]]


compare_collections

index_collection

sort_collection


foreach_in_collection my_cells [get_cells -hier * \
                 -filter "is_hierarchical == true"] {

echo "Instance [get_object_name $cell] is hierarchical"

}

  # Filtering operators: ==, !=, >, <, >=, <=, =~, !~
filter_collection [get_cells *] "ref_name =~ AN*"
get_cells * -filter "dont_touch == true"
get_clocks * -filter "period < 10"


  # List all cell attributes and redirect output to a file
redirect -file cell_attr \
          {list_attributes -application -class cell}

 # Grep the file for cell attributes starting with dont_
UNIX% grep dont_ cell_attr | more


  # List the value of the attribute dont_touch
get_attribute <cell_name> dont_touch


# Example: Identify glue cells in the current design
set GLUE_CELLS \
        [get_cells * -filter "is_hierarchical == false"]
```


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Compile Flow

```
# Enable multi-core optimization (Additional licenses may be required):
set_host_options -max_cores <#>
report_host_options
remove_host_options

# Prevent specific sub-designs from being ungrouped:
set_ungroup <top_level_and/or_pipelined_blocks> false
# Prevent DesignWare hierarchies from being ungrouped:
set_app_var compile_ultra_ungroup_dw false

# If needed: Disable boundary optimization on specific sub-designs:
set_boundary_optimization <cells or designs> false
# Allow only constant propagation
set_app_var compile_enable_constant_propagation_with_no_boundary_opt true
# Disable constant propagation on specific sub-block pins
set_compile_directives -constant_propagation false [get_pins "SUB2/In2 SUB2/In3"]

# If needed: Exclude specific cells/design from adaptive retiming (-retime)
set_dont_retime <cells_or_designs> true


# Manually controlling register replication:
set_register_replication [-num_copies 3 | -max_fanout 40] [get_cells B_reg]
# If needed: Change register replication naming style:
set_app_var register_replication_naming_style "%s_rep%d"
# To enable register replication across hierarchy:
set_app_var  compile_register_replication_across_hierarchy true

# Select the scan register style for test-ready synthesis, and disable automatic shift-register identification, if applicable:
set_scan_configuration -style <multiplexed_flip_flop | clocked_scan | lssd | aux_clock_lssd>
set_app_var compile_seqmap_identify_shift_registers false

# If needed: Maintain registered pipeline outputs if required
set_dont_retime [get_cells U_Pipeline/R12_reg*] true

# Enable register retiming if your design(s) contain(s) pure pipelines
set_optimize_registers true -design My_Pipeline_Subdesign -clock CLK1 -delay_threshold  <clock_period>

# Enable high effort timing optimization
set_app_var compile_timing_high_effort true

# Disable runtime intensive settings
compile_prefer_runtime

# Selectively disable/enable structural congestion optimization on sub-designs, which is performed  on the entire hierarchy, by default, with
# compile_ultra -spg
set_congestion_optimization [get_designs A] false

# Prioritize setup timing or max-delay over DRC
set_cost_priority -delay
# Disable DRC fixing on the entire clock network
set_auto_disable_drc_nets -on_clock_network true

# First compile: Enable/disable optimizations as needed.
# The -spg option is recommended once the floorplan definition is complete (2nd pass synthesis):
compile_ultra -scan -retime  [-spg]  [-no_boundary]  [-no_autoungroup]  [-no_design_rule]  [-no_seq_output_inversion]

# Apply more focus on violating critical paths, as necessary
group_path -name <group_name> -from <path_start> -to <path_end> -critical range <10% of max delay goal> -weight 5

# Perform an incremental ultra compile:
compile_ultra -scan -retime -incremental  -spg

# Final Area Recovery
optimize_netlist -area

# Parallel command execution using a maximum of 4 cores
update_timing
parallel_execute [list                  "report_constraint -all >  $mylogfile"  \
                                        "report_timing          >> $mylogfile"  \
                                        "report_area            >> $mylogfile"  \
                                        "report_qor             >> $mylogfile"]
report_congestion
```


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Net Layer Optimization

```
# There are 3 different ways to do net layer optimization


# 1a. Automatic net layer optimization using top 2 layers
compile_ultra –spg

# 1b. Enabling automatic net layer optimization using multiple layers
set_app_var spg_enable_zroute_layer_promotion true

compile_ultra –spg


# 2. Pattern based layer aware optimization
remove_net_search_pattern –all

set PATTERN_FO [create_net_search_pattern –fanout_lower_limit 150.00]
set_net_search_pattern_delay_estimation_options -min_layer_name M6 –max_layer_name M8 –pattern $PATTERN_FO

set PATTERN_NL [create_net_search_pattern -net_length_lower_limit 100]
set_net_search_pattern_delay_estimation_options -min_layer_name M7 -max_layer_name M8 -pattern $PATTERN_NL

set PATTERN_CRIT_IO [create_net_search_pattern -setup_slack_upper_limit "-1.0" -connect_to_port]
set_net_search_pattern_delay_estimation_options -min_layer M7 -max_layer M10 -pattern $PATTERN_CRIT_IO

compile_ultra –spg


# 3. Pattern based layer aware optimization
set_net_routing_layer_constraints –min_layer_name M5 -max_layer_name M6 [get_net top/sub/net1]

extract_rc –estimate ;# RC estimation
compile_ultra –incremental –spg ;# Optimization

# Binary handoff of Net Layer Optimization to ICC

# In ICC

read_ddc design.ddc
restore_spg_placement ;# To see DC-T placement in ICC
place_opt –spg


# ASCII handoff of Net Layer Optimization to ICC

# In DC-T
compile_ultra –spg
report_net_routing_layer_constraints [get_nets –hier] –output net_layer.tcl
write_file –format verilog –hierarchy –output design.v
start_icc_dp –f icc_dp.tcl; # generate the std_cell.def file
write_sdc

# In ICC
read_file –format verilog design.v
set_app_var spg_enable_ascii_flow true
read_def fp.def
read_def std_cells.def
read_sdc design.sdc
source net_layer.tcl
place_opt –spg
```

Application note available on SolvNet:
"How to Implement Net Layer Optimization With Design Compiler Graphical" https://solvnet.synopsys.com/retrieve/033546.html

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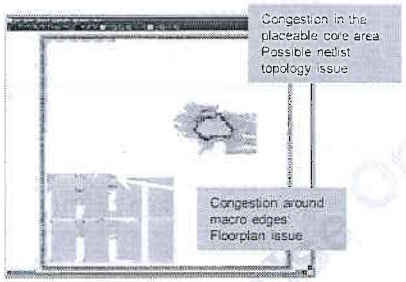
Congestion analysis

```
# Use same global and common Zroute options with DC and ICC
set_route_zrt_global_options ...
set_route_zrt_common_options ...
```

Congestion can be visually analyzed in the GUI:
`dc_shell-topo> start_gui`
Design Vision GUI: Windows → New Layout Window
Layout Window: View → Map Mode → Reload
Command equivalent: `report_congestion -build_map`

Congestion can be reported in text format:
`dc_shell-topo> report_congestion`



Overall Congestion

```
Both Dirs: Overflow = 3621
H routing: Overflow = 1724
V routing: Overflow = 1897
```

Congestion Direction | Overflow: Total # wires for which a GRC routing resource was not available

Worst Hotspot

Max = 13	(1 GRCs)
Max = 13	(1 GRCs)
Max = 8	(1 GRCs)

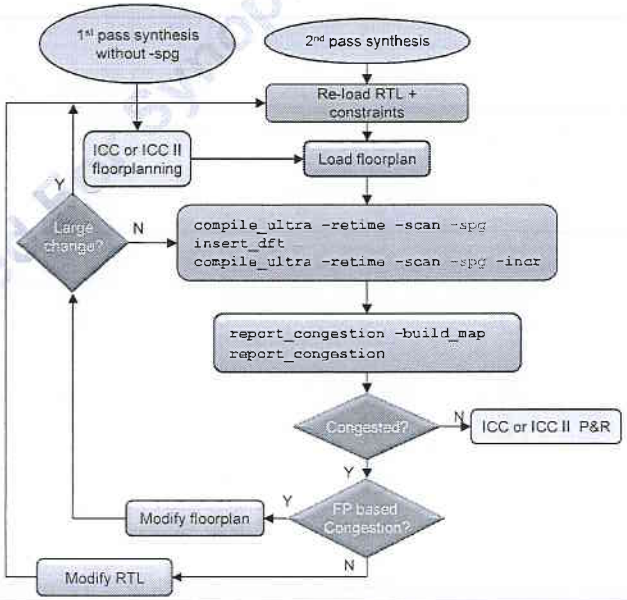
Maximum overflows for a single GRC | # GRCs with maximum overflow

Violating GRCs

GRCs = 2247	(2.73%)
GRCs = 1139	(1.38%)
GRCs = 1108	(1.34%)

GRCs with *any* overflow | Percentage Violating GRCs to Total GRCs

Synopsys Physical Guidance (SPG) flow



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